THE CAPITALISATION GRADIENT IN SEARCH-INDUCED OVERFITTING

A Controlled Study of Automated Strategy Search Across US Equities

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Abstract

A backtested trading result is systematically inflated when it is the single best outcome of a large and undisclosed search. Quantitative finance has established this in theory and supplied a correction for it, but the correction requires the number of trials as an input, and no prior study has measured how large the inflation is at the moment of selection or whether its severity depends on the kind of stock searched. This dissertation measures both under controlled experimental conditions.

An exhaustive, pre-specified search of a deliberately naive moving-average crossover rule was run across 2,808 valid parameter configurations for each of thirty US-listed equities, drawn in equal number from the large-, mid- and small-capitalisation tiers, over the ten-year window from January 2016 to December 2025, producing 84,240 valid trials. The final thirty per cent of each price series was sealed and never consulted during the search. Each selected strategy was then evaluated on that sealed data, corrected for the number of trials using the Deflated Sharpe Ratio, and compared against a buy-and-hold position in the same stock. The entire experiment was repeated with a relative strength index mean-reversion rule across a further 51,840 trials, giving 136,080 trials in total.

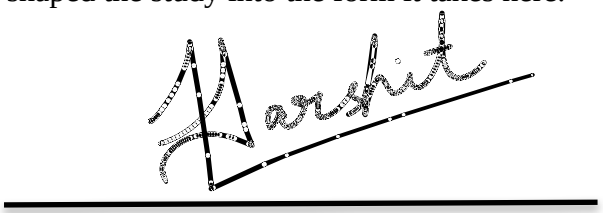
The method proved reliable as a generator of confident figures and unreliable as a source of skill. The mean reported in-sample Sharpe ratio of 0.76 fell to a realised 0.26 out of sample, and across all trials the correlation between reported and realised performance was only +0.17. Of the thirty selected strategies, nine lost money, only five survived the multiple-testing correction, only eight beat simply holding the stock, and none of the five achieved both in a meaningful sense. For eleven of the thirty, the reported figure fell below the level that a search of this size produces from data containing no signal at all. The central original finding is a capitalisation gradient: under the trend-following rule the gap between reported and realised performance widened monotonically from 0.22 in large-caps to 0.57 in mid-caps and 0.72 in small-caps, an ordering confirmed as statistically significant (Spearman rho = −0.44, p = 0.015), with the in-sample to out-of-sample correlation turning negative among small stocks. The robustness experiment qualified this: search-induced inflation reproduced under the opposite strategy family, the mean reported Sharpe of 0.90 falling to a realised 0.10, but the capitalisation gradient did not (rho = −0.20, p = 0.29), establishing the gradient as a feature of trend-following search rather than a universal law.

The study contributes a controlled measurement of an effect the literature had predicted but not quantified, and a five-step evaluation checklist through which an ordinary investor can judge a backtested claim without institutional resources.

Keywords: backtest overfitting; multiple testing; deflated Sharpe ratio; market capitalisation; automated strategy search; market efficiency.

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I am grateful to my supervisor, Professor Wasim Ahmed, for his guidance throughout this project, and in particular for the challenge to generate primary evidence rather than review the evidence of others, which shaped the study into the form it takes here.



Signature

List of Abbreviations and Key Terms

The terms below recur throughout the dissertation and are defined here so the reader can follow Chapters 1 to 6 without pausing. They are ordered by theme — the performance measures first, then the search and testing concepts, then the trading rules and study-specific terms.

Term	Definition
Sharpe ratio	Annualised return per unit of total volatility (both up and down swings). The main measure of risk-adjusted performance used throughout.
Sortino ratio	Annualised return per unit of downside volatility only. A check on the Sharpe ratio, since it does not penalise a strategy for upside swings.
Maximum drawdown (MaxDD)	The worst peak-to-trough loss over a period. A measure of whether a strategy would have been bearable to hold.
In-sample (IS)	The first seventy per cent of the price series, over which the parameter search was conducted.
Out-of-sample (OOS)	The sealed final thirty per cent of the price series, never consulted during the search. This is the genuine test of a strategy.
Trial	One evaluation of one parameter configuration on one stock. The study runs 2,808 trials per stock for the main rule.
Parameter (grid) search	Testing every combination of a rule's settings (e.g. moving-average lengths, stop-loss) and keeping the single best-performing one.
Overfitting	When a strategy fits the past data so closely that it captures its random noise rather than any real pattern, and so fails on new data.
Multiple testing	The problem that when many strategies are tried and only the best is reported, the best one looks good largely by chance, even if none has real skill.
Selection bias	The inflation in a reported figure caused by choosing the single best result out of many, rather than reporting a fair average.
Deflated Sharpe Ratio (DSR)	A correction that adjusts a reported Sharpe for the number of trials it was chosen from. It gives the probability that the strategy's true Sharpe is above zero. Below 0.95 is treated as indistinguishable from luck.
Buy-and-hold (B&H)	The return from simply owning the stock over the same window and taking no other

Term	Definition
	action. The benchmark every strategy must beat to count as skill.
Moving average (MA)	The primary trading rule examined, a moving-average crossover: a position opens when a fast average rises above a slow one and closes on the reverse.
Relative Strength Index (RSI)	The mean-reversion rule used as the robustness check, buying when a stock is oversold and selling when overbought.
Capitalisation tier	The size grouping of a stock: large-, mid- or small-capitalisation. The sample has ten stocks in each tier.
Capitalisation gradient	The study's central finding: the gap between reported and realised performance widens as stocks get smaller.
Basis points (bps)	One basis point is one hundredth of one per cent. The transaction cost here is ten basis points (0.10%) per trade.
rho (ρ)	The Spearman rank correlation coefficient, used to test whether the divergence is ordered across capitalisation tiers.

Chapter 1: Introduction

1.1 Background and Context

Investment decision-making has been progressively delegated to machines. Quantitative funds, algorithmic execution and factor-based portfolio construction have each moved part of the decision from the analyst to the model, and tools once reserved for institutions are now within reach of the ordinary investor. Software that will search thousands of trading configurations and report the single best one is freely available, and its output arrives wrapped in the language of rigour: a high Sharpe ratio, an impressive backtested return, a confident recommendation.

That arrangement creates an asymmetry. The investor sees the winning configuration but not the search that produced it, and so faces a question that is easy to state and hard to answer: can the reported figure be trusted? The question has become more pressing, not less, as the tools have improved. A retail platform that once offered a handful of indicators now offers automated optimisation across entire parameter spaces, and the newest systems built on large language models will propose, test and report strategies with no human specifying the search at all. The gap between what such a system reports and what it actually establishes is therefore no longer a technical curiosity confined to quantitative research desks; it is a practical problem for anyone deciding whether to act on a number.

1.2 The Problem, the Gap and the Contribution

Quantitative finance has known for decades that a reported backtest figure is systematically inflated when it is the best result of a large, undisclosed search. Bailey and López de Prado (2014) establish that where the number of configurations tested is not accounted for, out-of-sample disappointment is a predictable outcome rather than a surprising one. Yet this knowledge sits mostly in the technical literature and rarely reaches the investor who most needs it. The individual investor, without access to the institutional tools that might scrutinise a claim, is left to judge impressive numbers at face value, and is most exposed to precisely the stocks where those numbers are least trustworthy: the smaller, more volatile companies marketed as sources of exceptional growth.

The gap this study addresses can be stated plainly. The literature has established the theory of search-induced overfitting, has supplied competing corrections that all depend on knowing how many configurations were tried, and has demonstrated through large replication studies that the resulting decay is real and concentrated among small, illiquid stocks. What it has not done is measure that inflation directly. No prior study has run a search of a known and disclosed size over a deliberately naive rule, sealed the test data, and quantified how far the reported figure diverges from the realised one at the moment of selection. Nor has any study tested whether the size of that divergence depends on the capitalisation of the stock searched, although the size and liquidity literature clearly implies that it should.

The originality of this dissertation lies in closing that gap, and its contribution is twofold: a controlled measurement of search-induced inflation at the instant of selection, and the identification of a capitalisation gradient in that inflation. The problem is not, in the end, an argument to be settled by citation. It is an empirical claim that can be measured, and measuring it requires recreating in miniature, and under full experimental control, the exact process by which impressive backtest claims are generated.

1.3 Theoretical Foundations

Three established bodies of theory anchor the argument, each supplying one element of the reasoning. Chapter 2 develops each in turn.

Market efficiency. The efficient market hypothesis (Fama, 1970) holds that prices already reflect available information, so consistently earning risk-adjusted excess returns should be difficult. Grossman and Stiglitz (1980) refine this into a workable equilibrium in which a small, costly amount of mispricing persists as the reward for gathering information. This construct supplies the study's prior expectation: genuine edge is scarce, so an apparently successful trading rule demands explanation rather than acceptance.

Multiple testing and selection bias. The statistics of multiple testing explain how apparent skill is manufactured. When many candidate strategies are evaluated and only the best is reported, the reported figure is high by construction whether or not any real edge exists (Harvey, Liu and Zhu, 2016). Bailey and López de Prado (2014) convert this insight into a correction, the Deflated Sharpe Ratio, which adjusts a reported Sharpe for the number of trials from which it was selected. This construct supplies the study's central mechanism and its principal analytical instrument.

Market microstructure and liquidity. Microstructure theory explains why the effect should not be uniform across stocks. Illiquid securities trade less frequently, on thinner volume and with wider spreads, so their recorded price series carry more noise relative to genuine information, and illiquidity bears far more heavily on small firms than on large ones (Amihud, 2002). This construct supplies the study's moderating variable: if a search finds patterns in a price series, a noisier series should yield more spurious patterns.

Taken together the three constructs generate a single expectation. Skill is scarce; searching manufactures its appearance; and the manufacture should be easiest where the data are noisiest. The hypotheses below state that expectation in testable form.

1.4 Relevance to Business Analytics

Business analytics is concerned with extracting defensible conclusions from data, and its central professional obligation is not to build models but to know when a model's output can be trusted. Whether an analytical method genuinely performs, or merely appears to because of how it was evaluated, is the discipline's foundational question. It is posed here about automated strategy search in finance, but it could equally be

posed about any predictive system deployed on a high-stakes decision. The topic accordingly sits at the intersection of the programme's two strands: its subject matter is financial, while its method is analytical, involving the controlled generation of data, the measurement of a dependent variable against pre-specified benchmarks, and the identification of bias in reported results.

1.5 Aim, Objectives and Hypotheses

1.5.1 Aim

The aim of this research is to establish, under controlled experimental conditions, how reliably an automated strategy-search method performs when used to identify the best trading configuration for a stock, and to determine whether that reliability varies systematically across large-, mid- and small-capitalisation stocks.

1.5.2 Objectives

Five objectives serve that aim: to generate primary quantitative evidence by running an exhaustive, pre-specified search of a naive moving-average rule across a balanced sample of thirty US stocks over the period 2016 to 2025; to measure the gap between the reported in-sample performance of the selected strategy and its realised out-of-sample performance; to test whether any apparent success survives correction for the number of trials and comparison against a buy-and-hold benchmark; to determine whether the size of the search-induced illusion differs systematically across the three capitalisation tiers; and to establish whether the findings hold under a second, opposite strategy family, so distinguishing properties of automated search in general from properties of one particular rule.

1.5.3 Hypotheses

The study is framed as four hypotheses rather than research questions, since each states a directional expectation derived from the theory above and each is decided by a specific test rather than by discussion.

	Hypothesis	Derived from	Tested in
H1	An automated search reliably produces a high reported in-sample Sharpe ratio, but that figure diverges substantially from the performance the same configuration realises on sealed out-of-sample data.	Multiple testing and selection bias	4.2
H2	The size of that divergence increases as market capitalisation falls, being smallest in large-capitalisation stocks and largest in small-capitalisation stocks.	Microstructure and liquidity	4.3–4.4
H3	Once corrected for the number of trials and compared against a buy-and-hold position in the same stock, few selected strategies remain distinguishable from luck or superior to passive ownership.	Market efficiency; multiple testing	4.5–4.7
H4	Because the divergence is generated by the search rather than by the rule searched, it will reproduce when the same procedure is applied to a strategy family with opposite logic.	Multiple testing and selection bias	4.9

Table 1.1 The four hypotheses and where each is tested.

1.6 Summary of Findings

The experiment produced a clear and consistent answer, stated here so that the reader approaches what follows knowing what was found.

The search performed its mechanical task on every stock, returning a configuration with a mean reported in-sample Sharpe ratio of 0.76 a figure a practitioner would be pleased to present. That figure did not survive contact with new data. The mean realised out-of-sample Sharpe of the same strategies was 0.26, and across the full body of 84,240 trials the correlation between reported and realised performance was only +0.17. Of the thirty strategies a practitioner would have reported and acted upon, nine lost money outright, only five remained distinguishable from luck once the number of trials was accounted for, and only eight beat a simple buy-and-hold position in the same stock. Not one achieved both. A further computation sharpens the point: for eleven of the thirty stocks the reported figure fell below the best result that a search of this size would be expected to produce from data containing no signal at all.

The failure was not uniform. Under the trend-following rule it followed an orderly capitalisation gradient, the gap between reported and realised performance widening from 0.22 in large-caps to 0.57 in mid-caps and 0.72 in small-caps. Among small stocks the relationship between reported and realised performance turned negative, meaning a strong in-sample result there was not merely uninformative but actively misleading. Repeating the entire experiment with an opposite strategy family qualified this. The inflation reproduced fully, and more strongly, with not one of thirty strategies surviving correction; the gradient did not reproduce. The inflation is therefore a general property of automated search, while the way it varies with capitalisation is specific to trend-following rules.

Hypothesis	Verdict	Decisive evidence
H1 Reported and realised performance diverge	Supported	Mean Sharpe 0.76 → 0.26; all-trials correlation +0.17; Wilcoxon p = 0.0006
H2 The divergence widens as capitalisation falls	Supported for the trend-following rule; not supported for mean reversion	Drop 0.22 / 0.57 / 0.72 across tiers; Spearman rho = −0.44, p = 0.015; RSI rho = −0.20, p = 0.29
H3 Little survives correction and benchmarking	Supported, and more strongly than anticipated	5 of 30 survive deflation, 8 of 30 beat buy-and-hold, none do both; 11 of 30 fall below the luck benchmark
H4 The inflation belongs to the search, not the rule	Supported	Opposite rule: 0.90 → 0.10, correlation +0.21, 7 of 30 survive deflation

Table 1.2 Findings at a glance.

1.7 Structure of the Dissertation

Chapter 2 develops the three theoretical constructs into a critical review, comparing the competing methods available for detecting and correcting search-induced inflation rather than summarising studies in sequence, and closing by locating the gap from which the four hypotheses are derived. Chapter 3 builds the design backwards from those hypotheses, justifying each decision against the alternatives reported in the literature and stating the limitation each imposes. Chapter 4 applies the design and tests the hypotheses in turn, reaching a verdict on each. Chapter 5 interprets those verdicts, asking not only whether the findings match the theory but why the observed patterns arise, weighing alternative explanations, and separating statistical from economic and sample-specific significance. Chapter 6 draws the contribution together, states the implications for practitioners and academics, bounds the conclusions with the study's limitations, and identifies where the work should be extended.

Chapter 2: Literature Review

Chapter 1 named three constructs — market efficiency, multiple testing, and microstructure — and drew four hypotheses from them. This chapter tests how far the existing evidence supports each, and where it runs out. The review is comparative rather than sequential: where several methods address the same problem, they are set against one another and this study's choice justified by that comparison.

2.1 Scope and Approach

The problem sits at the intersection of five bodies of work: the theory of market efficiency and the empirical record on technical trading rules, which sets the baseline for how much genuine skill such a rule could hold; the statistics of multiple testing and backtest overfitting, which explains why searching many strategies inflates the best result; the methodological literature on validation design, which offers cross-validation and walk-forward alternatives to correction; the replication literature, which measures how far published findings decay out of sample; and the work on firm size, liquidity and efficiency, which explains why the least liquid stocks are most vulnerable to search-induced illusion.

The search covered the leading finance journals, including the *Journal of Finance*, the *Review of Financial Studies*, the *Journal of Financial Economics*, the *Journal of Portfolio Management* and the *Journal of Financial Data Science*, together with the statistical literature on multiple comparisons and recent work on machine learning in investing. Search terms combined backtest overfitting, multiple testing, data snooping, Sharpe ratio, deflated Sharpe, cross-validation, the size effect, liquidity and algorithmic trading. Seminal papers were located first and traced forward through citation, so that foundational concepts are paired with the work that updates them.

2.2 The Efficiency Baseline and the Record of Technical Trading

Any claim that an automated system can beat the market must be judged against a prior established more than fifty years ago. Fama’s (1970) efficient markets hypothesis holds that prices already reflect available information, so consistently earning risk-adjusted excess returns should be difficult. Grossman and Stiglitz (1980) refined this into a workable equilibrium: markets cannot be perfectly efficient, because if they were, no one would be paid to gather information; instead a small, costly amount of mispricing persists as the reward for the informed. The practical implication is the same for both. Genuine, repeatable edge is scarce, and most of what looks like skill is compensation for risk or simply noise.

Fama and French (2010) put a number on that scarcity. Studying the whole cross-section of US equity mutual funds, they show that the aggregate active portfolio sits close to the market, so before costs active management is a zero-sum game and after costs a negative-sum one. Using bootstrap simulations they find that very few funds earn enough to cover their costs, and that skilled and unskilled performance is almost

impossible to separate from luck except in the extreme tails. Their method is itself instructive here, because it prefigures this study's logic: rather than take reported alphas at face value, they construct the distribution of outcomes that chance alone would produce and ask whether the observed cross-section looks any different.

Two earlier contributions frame that debate more finely and are worth setting against Fama and French rather than merely listing alongside of them. Kosowski, Timmermann, Wermers and White (2006) apply a bootstrap procedure to the same question and conclude that a minority of managers do display genuine skill, particularly among growth funds, while Barras, Scaillet and Wermers (2010) apply a false discovery rate framework and find that the proportion of truly skilled managers declined sharply over their sample. The three studies use the same underlying data and reach different conclusions, and the difference lies almost entirely in the statistical machinery applied to separate skill from luck. That disagreement matters for this dissertation: it demonstrates that the correction chosen is not a neutral technical detail but a substantive determinant of what is concluded, which is why section 2.3 compares the available corrections rather than adopting one by default.

The record on technical trading rules specifically is more directly relevant, since the rule examined here belongs to that family. Brock, Lakonishok and LeBaron (1992) found that moving-average and trading-range rules produced returns on the Dow Jones Industrial Average that appeared to exceed a buy-and-hold benchmark. That finding did not survive scrutiny. Sullivan, Timmermann and White (1999) re-examined the same universe using White's (2000) reality check bootstrap, which explicitly accounts for the full universe of rules considered rather than only the one reported, and concluded that the apparent superiority largely disappears once data-snooping bias is corrected. Bajgrowicz and Scaillet (2012) reached a compatible conclusion using a false discovery rate approach, finding that after correction and realistic transaction costs few technical rules generate persistent profits.

This sequence matters in two ways. Substantively, it establishes that the moving-average crossover is not credited with genuine predictive edge in the modern literature, which is precisely the property required of a rule chosen to isolate the effect of search. Methodologically, it is a demonstration of the very mechanism this dissertation measures: a rule that appeared to work was shown not to, once the size of the search behind it was taken into account. Lo, Mamaysky and Wang (2000) offer the most sympathetic modern reading of technical analysis, finding through kernel regression that certain patterns carry modest incremental information; even that result is framed as informational rather than as an exploitable edge after costs.

The strength of this strand is its empirical weight and its equilibrium logic. Its limitation, for present purposes, is one of context rather than correctness. It was written about human fund managers making a modest number of discretionary bets, or about researchers testing a universe of rules they themselves enumerated. It does not address the situation in which a machine mechanically tests thousands of configurations in an afternoon and

reports only the single best one, without disclosing or, in the newest systems, without even recording how many were tried.

2.3 Multiple Testing and Backtest Overfitting: The Central Mechanism

If the efficiency literature establishes that luck dominates, the multiple-testing literature explains the exact mechanism by which luck is manufactured and then mistaken for skill.

Harvey, Liu and Zhu (2016) survey the hundreds of factors proposed to explain the cross-section of returns and argue that, given this scale of data mining, the conventional significance threshold is simply wrong. They propose that a newly claimed factor should clear a t-statistic of roughly 3.0 rather than the familiar 2.0, and conclude that most claimed findings in financial economics are probably false. Their contribution is to convert a vague worry about data mining into a concrete hurdle that scales with the number of tests performed. Harvey and Liu (2020) extend this into a formal framework for false and missed discoveries, showing that the correction must balance two errors rather than one, since an excessively severe threshold discards genuine findings alongside spurious ones.

Bailey and López de Prado (2014) turn the same insight into a tool aimed directly at the metric this study uses. They observe that modern quantitative teams run millions of analyses over a dataset in search of a profitable rule, so the best in-sample Sharpe ratio found is almost guaranteed to be high even when no real edge exists. Their Deflated Sharpe Ratio corrects a reported Sharpe for selection bias, the number of trials and non-normal returns, and returns the probability that the true Sharpe exceeds zero. The related work of Bailey, Borwein, López de Prado and Zhu (2014, 2017) develops the associated probability of backtest overfitting and demonstrates, strikingly, that with enough trials a researcher can achieve any desired in-sample Sharpe ratio on data that is pure noise.

Because several corrections are available and they do not agree, they are better compared than listed. The classical procedures address a different problem from the one at issue here. The Bonferroni adjustment divides the significance level by the number of tests and is the most conservative option available; it assumes independence between tests, which parameter configurations drawn from a single grid on a single price series manifestly are not, and at 2,808 trials it becomes so severe as to be uninformative. Holm's (1979) step-down procedure retains Bonferroni's control of the family-wise error rate while improving its power, but inherits the same independence difficulty. Benjamini and Hochberg (1995) reframe the objective entirely, controlling the expected proportion of false discoveries rather than the probability of any, which suits screening many candidate anomalies far better than evaluating a single strategy already selected. The bootstrap reality check of White (2000), refined by Hansen (2005) into a test for superior predictive ability, tests whether the best rule in a universe outperforms a benchmark once the whole universe is accounted for, and is arguably the most

rigorous option; it requires, however, the full return series of every candidate and substantial computation, and it too returns a verdict rather than a graded measure.

Set against these, the deflated Sharpe ratio has three properties that suit the present design specifically. It operates directly on the Sharpe ratio, which is the metric under examination, rather than requiring translation into a t-statistic. It takes the trial count as an explicit input, which makes this study's argument about disclosure operational rather than rhetorical. And it returns a probability rather than a pass-or-fail verdict, permitting stocks to be compared with one another and that comparison, rather than a binary count of survivors, is what allows a capitalisation gradient to be detected at all. The choice is therefore driven by what H2 requires, not by convenience.

The shared limitation of the whole strand is a single, load-bearing assumption: every one of these corrections presupposes a disciplined researcher who knows, records and can report how many strategies were tried. None anticipates a system that runs the search autonomously and either declines to disclose the trial count or, more fundamentally, does not itself track it. The correction exists and works; the one input it cannot function without is exactly the quantity the newest search technology removes from view. It means an entire class of modern performance claims arrives pre-immunised against the standard correction, not because the correction fails but because its denominator has gone missing.

2.4 Validation Design: Cross-Validation, Walk-Forward and Their Limits

Correction is not the only remedy available. A parallel literature argues that the problem should be addressed through the design of the validation itself rather than through an adjustment applied afterwards, and the two approaches are best understood as competing rather than complementary.

The simplest design-based remedy is the single chronological hold-out, in which a strategy is fitted on an initial portion of the series and tested on a sealed remainder. Its virtues are transparency and the absence of information leakage from future to past. Its weakness is that it produces exactly one out-of-sample estimate, itself subject to the conditions of the chosen window, and that a researcher free to reselect the split point can quietly reintroduce the selection problem the design was meant to remove.

Walk-forward analysis addresses the first weakness by rolling the fitting and testing windows forward repeatedly, generating a sequence of out-of-sample estimates rather than one. It reflects how a strategy would actually be redeployed over time and is standard in practitioner settings. It does not, however, solve the underlying problem, because the number of configurations tested at each step remains uncorrected, and repeated re-optimisation can itself become a source of overfitting.

López de Prado (2018) develops the most sophisticated design-based alternative, combinatorially symmetric cross-validation, which generates many train-test splits from the same series and produces a distribution of

out-of-sample outcomes rather than a point estimate; from that distribution the probability of backtest overfitting can be computed directly. He also introduces purging and embargoing to prevent leakage across adjacent observations. The approach is powerful and, in principle, more informative than a single hold-out.

Two considerations nonetheless favour the single sealed split here. The first is that cross-validation of any kind reuses the same data repeatedly, and a strategy selected on the basis of an average across many splits has, in a meaningful sense, seen all of the data. Where the object of study is precisely the divergence between what a search reports and what genuinely unseen data delivers which is what H1 asserts only data that has never been touched can serve as the test. The second is interpretability: a single chronological split reproduces exactly what a practitioner does when fitting on history and deploying forward, and the study's purpose is to measure what happens to that practice rather than to demonstrate a superior one. The cost, that the out-of-sample estimate depends on one particular window, is acknowledged in Chapter 3 and revisited in the limitations.

2.5 Replication and Post-Publication Decay: The Problem Is Real

A sceptic might argue that overfitting is a theoretical concern that does not much affect real strategies. Several large replication studies close that escape route with evidence.

Hou, Xue and Zhang (2020) re-test 452 published anomalies under consistent, conservative standards. They report that 65 per cent fail to clear even a single-test hurdle, and that imposing the higher multiple-testing hurdle raises the failure rate to 82 per cent, with surviving effects far smaller than first claimed. They attribute the failures to method rather than bad faith: much of the original evidence, they show, was driven by microcap stocks, which exert outsized influence in equal-weighted tests. Once microcaps are controlled with appropriate breakpoints and value-weighting, the bulk of the anomalies dissolve.

That conclusion has itself been contested, and the disagreement is instructive rather than incidental. Jensen, Kelly and Pedersen (2023) apply a hierarchical Bayesian framework to a large global dataset and argue that the replication rate is considerably higher than Hou, Xue and Zhang report, contending that the apparent crisis is partly an artefact of testing each factor in isolation rather than jointly. Chen and Zimmermann (2022) likewise find, from an open-source reconstruction of published predictors, that most anomalies do replicate in-sample, though with attenuated magnitudes. The literature therefore does not speak with one voice on how severe the replication problem is, and a review presenting only the pessimistic finding would misrepresent the evidence. What is not contested, and what matters here, is that in-sample magnitudes systematically overstate what follows.

McLean and Pontiff (2016) approach from a different direction, tracking 97 predictors after their original samples end. They find portfolio returns 26 per cent lower out-of-sample and 58 per cent lower after publication, and that the decay is steepest for the predictors with the highest in-sample returns. This last detail

bears directly on H1: it is empirical evidence that the more impressive the in-sample figure, the more of it is illusory. They also document that after publication predictor portfolios become more correlated and concentrate the surviving return in the stocks hardest and costliest to trade, consistent with sophisticated investors learning about the mispricing from the research itself. Chordia, Goyal and Saretto (2020) reinforce the statistical side, generating over two million trading signals from accounting data and showing that the number surviving appropriate correction is close to what chance alone would produce.

The distinction McLean and Pontiff draw, between decay caused by data mining and decay caused by real-world arbitrage, is one this study can sidestep entirely, because its out-of-sample window is sealed and the strategies are never published or traded; any gap measured is pure selection effect. That is also the strand's limitation for present purposes. Both papers measure human-authored, fully disclosed research decaying gradually over time. Neither isolates, under controlled conditions and with the search as the sole possible cause, how much a single automated search inflates a reported figure at the instant of selection.

2.6 Firm Size, Liquidity and the Efficiency of Small Stocks

The replication literature contains a detail that points directly to H2. When Hou, Xue and Zhang trace the failure of published anomalies to their source, they find much of the original evidence was driven by the smallest, least liquid stocks. Vulnerability to false discovery is therefore not uniform across the market but concentrated among small stocks.

The foundational result is Banz (1981), who found that smaller firms had earned higher risk-adjusted returns than larger ones, a regularity he read as evidence that the capital asset pricing model was misspecified. For the present study the enduring value of that work lies less in the return premium, which later research has contested, than in the more basic and uncontested point it established: that small and large stocks are not statistically interchangeable. Fama and French (1993) later formalised size as one of the systematic factors needed to describe the cross-section of returns.

A critical reading must acknowledge that the size premium itself has become contested, and the contest is genuine rather than marginal. Asness, Frazzini, Israel, Moskowitz and Pedersen (2018) argue that the premium is weak in raw form but reappears robustly once quality is controlled for, contending that the apparent weakness reflects the concentration of junk among small firms. Linnainmaa and Roberts (2018) go further, showing that many accounting-based anomalies including size perform substantially worse in pre-sample data, which they interpret as evidence of data mining in the original discoveries. This debate does not weaken the strand's relevance, because H2 does not rely on a size premium and makes no claim that small stocks earn more. It relies instead on a different and undisputed property, one that even the sceptics of the premium accept: that small-capitalisation stocks are noisier, less liquid and less efficiently priced than large ones.

The mechanism linking size to inefficiency is liquidity. Amihud (2002) shows that expected market illiquidity is positively associated with expected stock excess return and, crucially, that illiquidity affects small-firm stocks far more strongly than large ones. Illiquid stocks trade less frequently, on thinner volume, with wider spreads and larger price impact per trade, so their recorded price series contain more noise relative to genuine information. Earlier work from Amihud and Mendelson (1986) and Roll (1984) had already observed that the prices of small firms are set in thinner, less continuously traded markets, so their measured returns are noisier and their estimated risk parameters less reliable. Pástor and Stambaugh (2003) establish that liquidity is a priced systematic risk in its own right. Novy-Marx and Velikov (2016) add the practical dimension that matters most for a trading study: after realistic transaction costs many anomaly strategies lose the greater part of their apparent profitability, and the erosion is most severe among small and illiquid stocks where turnover is costly.

Assembled, this strand supplies the theoretical foundation for H2. A strategy search operates by finding patterns in a price series; a noisier series contains more chance patterns that fit the past without predicting the future. If small stocks are noisier and less efficiently priced than large ones, an identical search applied to small stocks should find more spurious patterns and produce more inflated in-sample results. The limitation of this strand is that it was developed to explain returns and risk in passive holding, not to explain how much an automated search inflates reported performance. It predicts, but has never directly measured, that search-induced overfitting should follow a capitalisation gradient.

2.7 Automation, Machine Learning and Language Models in Investment Search

A final strand situates the study in the present. The search this dissertation conducts in a controlled and countable way is increasingly performed by automated systems at a scale no human could match.

Gu, Kelly and Xiu (2020) show that machine-learning methods can add genuine economic value in measuring risk premia when applied with discipline: a clear out-of-sample protocol, a large dataset and honest reporting. Their work establishes an important boundary, namely that automation is not inherently suspect and that a search can be trustworthy if the protocol is. Arnott, Harvey and Markowitz (2019) codify what that discipline requires: pre-specification, sealed out-of-sample data, scepticism of complexity and, above all, disclosure of the extent of the search. Welch and Goyal (2008) provide a cautionary counterpoint from an adjacent literature, demonstrating that a long list of return predictors performing well in-sample failed to outperform a simple historical mean out-of-sample a finding that anticipates, in a different setting, exactly the divergence H1 asserts.

The concern motivating the wider debate is that the most recent automated and agentic systems increasingly abandon these safeguards. Systems built on large language models will now propose, test and report trading strategies with the search conducted internally and unrecorded. Lopez-Lira and Tang (2023) demonstrate that

such models can extract return-predictive signal from news headlines, establishing that the capability is real rather than speculative. What is absent is any disclosure discipline: these systems report headline figures while disclosing neither the number of strategy variants explored nor the conditions under which they were tested. In the terms of section 2.3, these are performance figures that cannot be deflated, because the trial count the correction requires is not available. This study does not test such a system directly; what it offers the debate is a controlled measurement of how large the search-induced illusion is, against which undisclosed claims can later be judged.

2.8 Synthesis: The Gap and the Hypotheses It Generates

Read as a chain, the five strands make one cumulative argument. Genuine skill is scarce and hard to tell from luck, and the specific family of rules examined here has repeatedly lost its apparent edge once the search behind it was accounted for (Fama and French, 2010; Sullivan, Timmermann and White, 1999). The mechanism that manufactures false skill is multiple testing, and every available correction requires knowing how many strategies were tried (Harvey, Liu and Zhu, 2016; Bailey and López de Prado, 2014). Design-based alternatives mitigate but do not remove the problem, and reuse data in ways a study of genuine out-of-sample divergence cannot accept (López de Prado, 2018). That the problem is real is confirmed by replication studies showing the most impressive in-sample figures decay the most and that failures concentrate in the least liquid stocks (Hou, Xue and Zhang, 2020; McLean and Pontiff, 2016), and the size and liquidity literature explains why (Banz, 1981; Amihud, 2002; Novy-Marx and Velikov, 2016).

What no study has done is measure, under controlled conditions with a deliberately naive rule, how large the inflation is at the moment of selection (H1), whether it follows the capitalisation gradient the microstructure literature implies but has never tested for automated search (H2), how much of it survives correction and realistic benchmarking applied together (H3), and whether it is a property of search itself rather than of the rule searched (H4). Prior work either theorises the mechanism, documents its aftermath, or explains the conditions that should aggravate it; none isolates and quantifies it in a single reproducible experiment designed so that the search is the only possible explanation for any in-sample success. Chapter 3 sets out the design that closes that gap.

The relationships reviewed above can be drawn together as a single causal chain, against which each hypothesis tests one link (Figure 2.8).

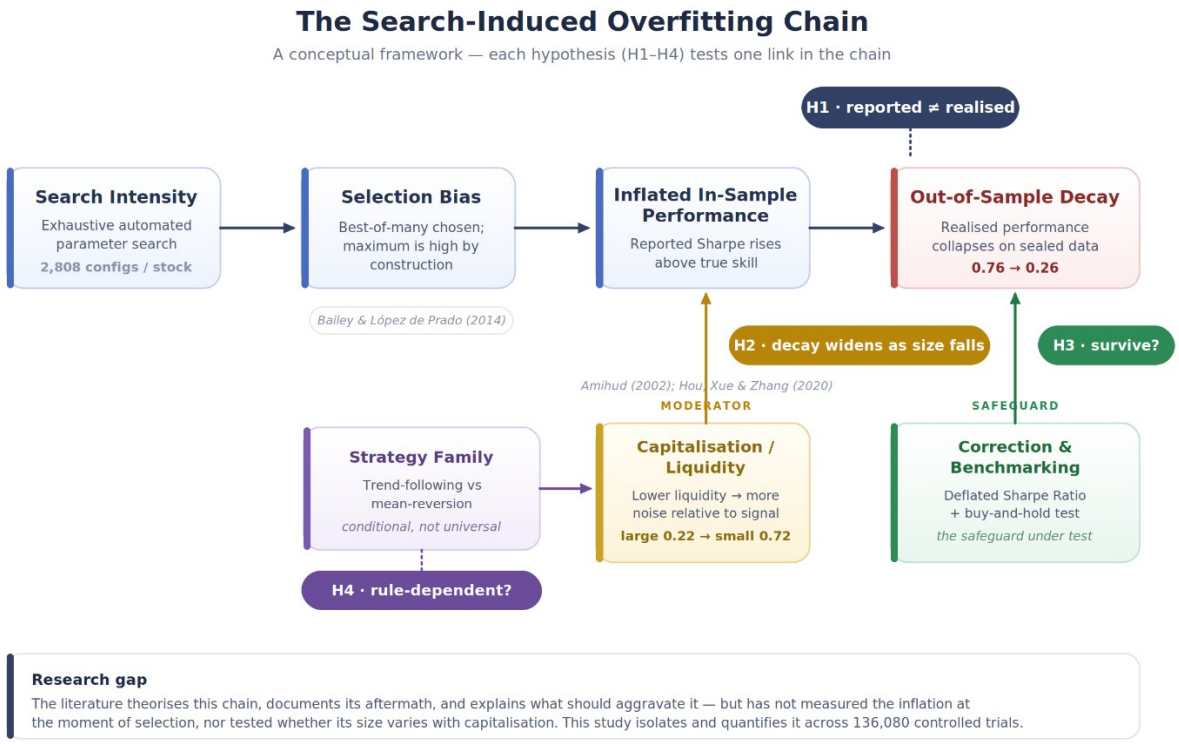


Figure 2.8 Conceptual framework: the search-induced overfitting chain.

H1 tests whether the inflated in-sample figure decays out of sample; H2 tests capitalisation as a moderator of that decay; H4 tests whether the moderation is itself conditional on strategy family; and H3 tests whether the standard correction catches what the chain produces.

Chapter 3: Research Methodology

Chapter 2 showed that the literature explains why search inflates a reported figure and supplies competing corrections, but has never measured that inflation directly or tested whether it varies with capitalisation. Closing that gap requires an experiment, not a review. This chapter builds the design backwards from the four hypotheses, stating for each decision the alternatives considered, the reason for the choice, and the limitation it imposes.

3.1 What the Hypotheses Require of the Design

Each hypothesis constrains the design, and taken together they determine it. Measuring the divergence between reported and realised performance (H1) requires the number of configurations searched to be known exactly, which rules out any approach relying on figures reported by others. Testing whether that divergence widens as capitalisation falls (H2) requires tiers of equal size, since unbalanced tiers confound capitalisation with sample size, and requires a correction that returns a graded probability rather than a verdict, since an ordering cannot be detected from a binary count of survivors. Testing whether apparent success survives correction and benchmarking (H3) requires the evaluation apparatus to be fixed before any result is seen, so that failures are exposed by design rather than discovered afterwards. Testing whether the effect belongs to the search rather than the rule (H4) requires the entire experiment to be repeated with a rule whose logic is opposite.

3.2 Philosophy and Strategy

The study adopts a positivist philosophy and a deductive, quantitative-experimental strategy, treating the relationship between search intensity and reported performance as an objective regularity measurable through controlled, replicable procedure. An interpretivist alternative, investigating through interviews how investors come to trust performance claims, would explain why misleading figures are believed but could not measure how misleading they are. The decisive argument is internal to the subject: the failure under examination is one of undisclosed, unrepeatable search, so a philosophy demanding pre-specification and replication is consistent with the study's own argument.

Three strategies were rejected. Analysing already-published performance figures would inherit the very flaw the study exposes, since their trial counts are unknown. Surveying practitioners would answer a behavioural question, not a measurement one. Searching only synthetic data would give perfect control but could not test H2, which concerns real differences between real classes of stock; that logic is retained as the validation step in section 3.10. The cost of the choice is that the findings concern a simple rule rather than a commercial system, accepted because the aim is to measure a mechanism, not to build a profitable strategy.

3.3 The Research Instrument

The instrument is a computational search framework applying a naive moving-average crossover rule: a long position opens when a fast average rises above a slow one and closes on the reverse crossover, subject to a minimum holding period and a stop-loss. Momentum, breakout and machine-learning rules were rejected for one reason. A rule that might contain genuine predictive power confounds the measurement, since in-sample success could then be attributed either to the search or to the rule. The crossover is chosen because it is naive and, as section 2.2 established, is not credited with genuine edge once data-snooping is corrected for, which makes attribution unambiguous. It is also among the rules most widely offered in retail software, so the demonstration speaks to the tools the study's audience actually encounters.

The framework searches a grid fixed before any analysis and never altered:

Parameter	Values	Count
Fast average (days)	3, 5, 8, 10, 12, 15, 20, 25	8
Slow average (days)	20, 30, 40, 50, 60, 80, 100, 150, 200, 250	10
Minimum holding period (days)	3, 5, 10, 15, 20, 30	6
Stop-loss	2, 3, 5, 8, 10, 15 per cent	6
Combinations	8 × 10 × 6 × 6	2,880
Degenerate (fast not shorter than slow)	excluded by design	72
Valid configurations per stock		2,808

Table 3.1 Moving-average parameter grid, fixed in advance and reproduced in full in Appendix A.

A transaction cost of ten basis points is charged on entry and exit and never tuned, so costs cannot become a hidden optimisation parameter. Publishing the grid is not a formality: the correction applied later requires the trial count as an input, and the study's argument is that undisclosed search removes precisely that input from view.

3.4 The Robustness Experiment

A single rule cannot distinguish a property of automated search from a property of that rule, which is what H4 asserts. The design therefore repeats the experiment with an opposite family: a relative strength index mean-reversion rule, buying when a stock is oversold and selling when overbought. Where the crossover is trend-following, this is contrarian, so a finding reproducing across both is unlikely to be an artefact of either. The grid searches lookback period across eight values, entry threshold across six, exit threshold across six and stop-loss across six, giving 1,728 configurations per stock and 51,840 further trials. It was sized to give a search comparable in magnitude to the primary experiment, and the values were fixed before any result was examined. The same stocks, window, split, cost and evaluation apparatus were used throughout, giving 136,080 trials in all.

3.5 Sample and Data

Thirty US-listed equities with a continuous ten-year history were drawn, ten per tier. An entire index would have given a larger sample but contains no small-caps and is itself a survivorship filter; a random draw would not have produced balanced tiers; multiple markets were rejected on tractability, each stock requiring 4,536 backtests. Three criteria were applied before any analysis: complete price history, equal tiers, and inclusion of stocks that fell as well as rose. The third is the survivorship control, since selecting only stocks that prospered would itself be an instance of the distortion under study and would flatter any long-biased rule. Thirty stocks is modest adequate to detect an ordering across three tiers but limiting the precision of tier estimates.

Daily prices for 1 January 2016 to 31 December 2025 were collected from Yahoo Finance through the yfinance interface, adjusted for splits and dividends. CRSP and Bloomberg are the professional standards and were unavailable; against that constraint Yahoo offers two properties suiting this design, being freely reproducible without subscription and being the source most likely to underlie the retail tools whose claims the study evaluates. Its limitations are that it excludes delisted securities, recalculates adjusted prices as dividends accrue, and documents corporate actions less fully than CRSP. None affects the internal comparison, since every tier is drawn from the same source, but each bounds the external claim. Collection was automated, removing transcription error. Each series was split chronologically, the first seventy per cent searched and the final thirty per cent sealed, preserving the time ordering essential to a realistic test.

3.6 Variables

The dependent variable is out-of-sample risk-adjusted performance, measured principally by the out-of-sample Sharpe ratio of the selected configuration. Two independent variables are examined: the reported in-sample Sharpe, representing the figure a practitioner would disclose, and the capitalisation tier, the moderating variable through which H2 is tested. The Sortino ratio, maximum drawdown, deflated probability and buy-and-hold benchmark are supporting measures and evaluation criteria, not variables a distinction maintained explicitly in section 4.1. Transaction costs are a controlled constant.

3.7 Analysis Plan and the Choice of Instruments

The sequence was specified before the data were examined. Three metrics are computed for every trial on both portions. For each stock the configuration with the highest in-sample Sharpe is selected, exactly as a practitioner reporting a backtest would, and its out-of-sample performance becomes the realised figure. Every other trial is retained, which is what makes the all-trials analysis possible. No trial was lost to missing data or computational failure, so completion of both valid grids was total.

Each instrument was chosen over specific alternatives:

Instrument	Alternatives rejected	Reason for the choice
Deflated Sharpe Ratio	Bonferroni; Holm step-down; Benjamini-Hochberg false discovery rate; White reality check	Operates directly on the Sharpe; takes the trial count as an explicit input; returns a graded probability rather than a verdict, which is what permits tiers to be compared and H2 to be tested at all. Bonferroni and Holm assume independent tests, which grid configurations are not; the reality check requires the full return series of every candidate.
Buy-and-hold benchmark	Index or equal-weight portfolio; momentum factor; risk-free rate	The most conservative, like-for-like test, holding the asset constant. A portfolio benchmark answers a different question; a momentum factor reintroduces the search problem; the risk-free rate sits inside the Sharpe already.
Single sealed chronological split	Walk-forward; combinatorially symmetric cross-validation	Cross-validation reuses the same data, so a strategy selected across many splits has in effect seen all of it. Only untouched data can measure the divergence H1 asserts.
Wilcoxon signed-rank, with paired t-test	Either alone	Paired data, no normality assumed. Agreement between a parametric and a non-parametric test on thirty pairs is stronger than either alone.
Kruskal-Wallis	One-way ANOVA	Tier performance cannot be assumed normal and variances differ across tiers.
Spearman rank correlation	Pearson	Kruskal-Wallis detects difference but not order, and H2 is a claim about ordering. Tier is ordinal, and a rank measure resists the extreme cases the sample contains by design.

Table 3.2 Analytical instruments and the alternatives considered.

3.8 Robustness and Validity

Five threats to validity were anticipated. Because every trial was retained, three can be measured rather than merely discussed.

Parameter sensitivity. No single parameter dominates: mean in-sample Sharpe varies by 0.018 across the fast average and 0.092 across the slow, with holding period and stop-loss contributing 0.034 and 0.033. More revealing, the winning configurations scatter across the whole grid, using all eight fast values, seven of ten slow values, all six holding periods and all six stop-loss levels. Were the rule to contain genuine structure the winners would cluster; that they do not is direct evidence the search fits noise particular to each series. The selection effect is quantifiable: the best in-sample Sharpe averages 0.764 against a median of 0.282 on the same stock, so selecting the maximum adds 0.48 before any question of genuine performance arises.

Look-ahead bias. The split is chronological, the sealed portion is never consulted during selection, and the grid was fixed in advance. One implementation detail cuts the other way and is disclosed: the out-of-sample window is evaluated as an independent series, so moving averages restart at its beginning and a configuration sits in cash until its slow average becomes available. With a median winning slow average of thirty days this costs about four per cent of a 754-day window, though for the six stocks whose winner used 150 days or more it reaches roughly a third. It suppresses realised return slightly and flatters drawdown, which reinforces rather

than undermines the finding of section 4.4 that milder drawdowns arose from absence from the market rather than skill in timing.

Survivorship bias. Addressed in sample construction by requiring stocks that fell as well as rose. The residual exposure lies in the data source, which excludes delisted securities, so companies that failed outright could not enter. This understates rather than overstates the effect, since a strategy searched on a stock that later delisted would fail more dramatically; the bias runs against the study's own finding.

Transaction costs. Fixed at ten basis points by design, since a tuned cost assumption would become a hidden optimisation parameter. Sensitivity to the assumption is therefore unexplored, and a flat rate does not capture spread, market impact and slippage, all larger for small-caps, as Novy-Marx and Velikov (2016) document. Because friction is greatest precisely where results are worst, a more realistic model would deepen the gradient rather than flatten it, so the finding is conservative.

Generalisability. One decade containing a prolonged rising market, one national market, thirty stocks, two strategy families. The benchmark comparison controls for market direction within each stock, and the robustness experiment establishes which findings survive beyond a single rule; the remaining boundaries are revisited in Chapter 6.

3.9 Assumptions in the Implementation

Four assumptions are embedded in the analysis and disclosed so the results can be read against them. The Sortino ratio uses the standard deviation of negative returns rather than the root mean square of below-target returns, a recognised variant returning slightly higher values; since it compares tiers rather than an external threshold, the comparison is unaffected. The deflated Sharpe omits the skewness and kurtosis adjustment, assuming approximately normal returns, which given the negative skew of strategy returns makes the correction slightly less severe than the full formulation, so the count of survivors is if anything generous. The standard deviation of trial Sharpe ratios required by the deflation formula was set to a fixed value of 0.30 across all stocks rather than estimated separately for each, a simplifying constant applied uniformly to both strategy families; per-stock estimation of this quantity is a sensitivity the present study does not explore, and is noted among the limitations in Chapter 6. The stop-loss closes a position but imposes no waiting period, so a configuration may re-enter next day if the crossover still holds. The relative strength index uses a simple rolling mean rather than Wilder's smoothing, applied consistently.

3.10 Validation and Ethics

Before application to real prices the framework was validated on thirty synthetic random-walk series of 2,514 days each, matching the length of the real data, and searched across the same 2,808-configuration grid. These series contain no signal by construction. The search nonetheless returned a mean in-sample Sharpe of 0.70,

which fell to a mean realised out-of-sample Sharpe of 0.17, with twelve of the thirty series returning a negative out-of-sample Sharpe despite containing no genuine edge of any kind. A framework reporting skill on data known to contain none would be unusable; one reproducing this expected collapse can be trusted to measure the same effect on real prices.

The study uses only public price data and involves no human participants, so no ethical approval was required. Two points deserve statement. The strategies demonstrate a statistical mechanism and are not investment advice; the conclusion is a caution against trusting exactly this kind of result. And the analysis is reported transparently, with the trial count disclosed, the grids published and every trial retained, in deliberate contrast to the undisclosed search the study identifies as the problem.

3.11 Summary

The design isolates the number of configurations searched as the single variable capable of explaining in-sample success, and measures the divergence between reported and realised performance as the outcome. A naive rule makes attribution unambiguous; an opposite second rule separates general findings from rule-specific ones; a balanced, survivorship-aware sample permits the cross-tier comparison; a sealed chronological split provides the honest test; deflation and benchmarking provide the correction. Chapter 4 applies this design and takes the four hypotheses in turn.

Chapter 4: Results and Analysis

This chapter applies the design specified in Chapter 3 to thirty US equities, drawn in equal number from the three capitalisation tiers, over the ten-year period from January 2016 to December 2025, and tests each hypothesis against the evidence. Sections 4.2 to 4.9 take H1 to H4 in order; section 4.11 records the verdicts.

Throughout, the chapter follows a single discipline. No reported figure is accepted at face value; every apparent success is tested against what it was measured on, corrected for how it was selected, and compared against what an uninformed alternative would have achieved. It is this discipline, rather than the raw numbers, that turns a set of results into an analysis.

4.1 Measures and How to Read Them

The variables were defined in section 3.6: out-of-sample risk-adjusted performance is the dependent variable, the reported in-sample Sharpe and the capitalisation tier are the independent variables. The quantities set out here are the measures through which those variables are observed and the criteria against which they are judged; they are not themselves variables.

The Sharpe ratio measures annualised return per unit of total risk and is the primary measure of the dependent variable. The Sortino ratio penalises only downside volatility, and is used to check whether apparent performance survives when the reward for lucky upside swings is removed. The maximum drawdown, the worst peak-to-trough loss, measures whether a strategy would have been survivable to hold rather than merely attractive on paper. Each is computed twice: in-sample over the first seventy per cent of the data, over which the search was conducted, and out-of-sample over the sealed final thirty per cent, which constitutes the genuine test. Two evaluation criteria complete the apparatus. The deflated probability is the likelihood, once the number of trials has been accounted for, that a strategy's true Sharpe exceeds zero, with a value below 0.95 marking a result indistinguishable from luck. The buy-and-hold benchmark is the performance of simply owning the stock over the same window.

4.2 Testing H1: Reported Against Realised Performance

The method performs its intended function: for every stock it searches 2,808 valid configurations and returns the one with the highest in-sample Sharpe. Across all thirty stocks the mean in-sample Sharpe of these selected configurations is +0.76, a figure a practitioner would be pleased to report. The critical question is whether it survives contact with new data. It does not: the mean out-of-sample Sharpe of the same strategies is +0.26, a fall of one half.

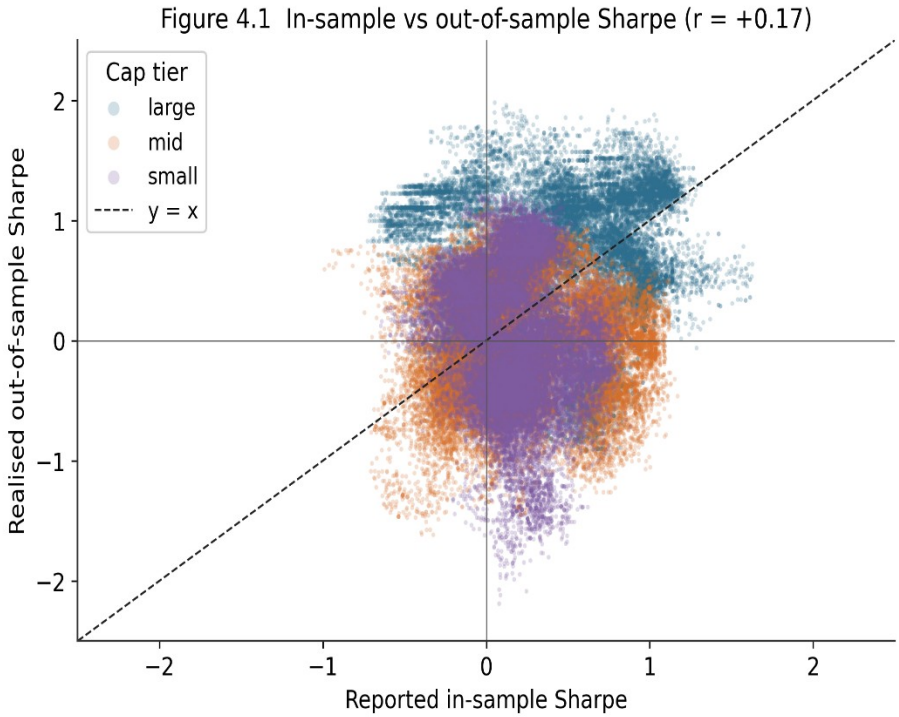


Figure 4.1 In-sample versus out-of-sample Sharpe for all 84,240 trials, coloured by capitalisation tier. A strong in-sample result predicts little about out-of-sample performance (r = +0.17). Source: author's analysis.

That fall is not simply bad luck in the selection, as Figure 4.1 shows. Across the full body of 84,240 trials the correlation between reported and realised performance is +0.17. Had the rule contained genuine structure, the cloud would concentrate along the diagonal, since configurations that worked on the past would tend to work on the future. Instead it is broad and largely uninformative: a strong in-sample result carries little reliable information about what follows. The selected strategy is not merely an unlucky draw from a good distribution; it is the maximum of a distribution that carries almost no signal.

The consequence is best seen not in the cloud but in the thirty strategies a practitioner would actually report and act upon. Nine of the thirty lost money outright out of sample. Only five remained distinguishable from luck once the number of trials was accounted for. And most tellingly, only eight beat a simple buy-and-hold position in the same stock meaning that in twenty-two of thirty cases an investor would have done better holding the stock and running no strategy at all. The method reliably produces a confident-looking number; it does not reliably produce a result that means anything.

H1 is supported.

The full set of selected strategies, grouped by tier, is reported in Table 4.1. It is the evidential basis for every claim in this chapter, and the individual cases discussed below can be traced through it.

Stock	Tier	IS Sharpe	OOS Sharpe	OOS Sortino	OOS MaxDD	Defl. p	Sig?	Beat B&H?
AAPL	large	1.63	0.63	0.70	−26%	0.99	Yes	No
IBM	large	0.61	1.48	1.79	−19%	0.29	No	Yes
INTC	large	0.33	0.10	0.13	−51%	0.34	No	No
JNJ	large	0.81	0.51	0.68	−22%	0.70	No	No
KO	large	0.61	0.20	0.23	−15%	0.42	No	No
META	large	0.72	1.11	1.60	−24%	0.59	No	No
MSFT	large	1.14	0.49	0.50	−22%	0.99	Yes	No
NFLX	large	0.92	1.21	1.49	−24%	0.77	No	No
NVDA	large	1.31	1.30	1.86	−42%	0.99	Yes	No
PG	large	1.00	−0.11	−0.10	−20%	0.89	No	No
DDD	mid	0.80	−0.14	−0.16	−79%	0.86	No	Yes
ETSY	mid	1.17	−0.46	−0.38	−50%	0.97	Yes	No
FOSL	mid	0.41	0.74	1.02	−52%	0.07	No	Yes
FTNT	mid	1.10	0.49	0.44	−30%	0.90	No	No
GNRC	mid	1.12	0.14	0.15	−31%	0.99	Yes	No
GPRO	mid	0.50	0.44	0.50	−54%	0.31	No	Yes
M	mid	0.68	0.71	0.78	−30%	0.42	No	Yes
UAA	mid	0.40	0.28	0.31	−35%	0.16	No	Yes
WHR	mid	0.41	−0.33	−0.30	−51%	0.28	No	No
YELP	mid	0.56	−0.39	−0.42	−44%	0.41	No	No
CALX	small	0.92	0.21	0.23	−29%	0.91	No	Yes
CENX	small	0.66	0.66	0.88	−53%	0.70	No	No
CRUS	small	0.64	0.44	0.48	−31%	0.57	No	No
EGHT	small	0.62	−1.39	−1.00	−87%	0.52	No	No
HELE	small	0.88	−0.72	−0.58	−73%	0.75	No	No
IIIN	small	0.52	−0.51	−0.43	−49%	0.57	No	No
MTH	small	0.86	0.09	0.11	−48%	0.72	No	No
PLAB	small	0.25	0.32	0.38	−62%	0.33	No	No
SCHL	small	0.74	−0.49	−0.49	−68%	0.30	No	No
SHOO	small	0.60	0.86	1.17	−33%	0.52	No	Yes

Table 4.1 Selected strategy for each of the thirty stocks, with reported and realised performance, downside risk, deflation and benchmark outcomes. Only five strategies are significant after deflation; only eight beat buy-and-hold; none of the five do both. Source: author's analysis.

4.3 Testing H2: The Capitalisation Gradient

H2 predicts that the divergence measured above is not uniform but widens as capitalisation falls. Under the trend-following rule it does, and the ordering is the study's central original finding.

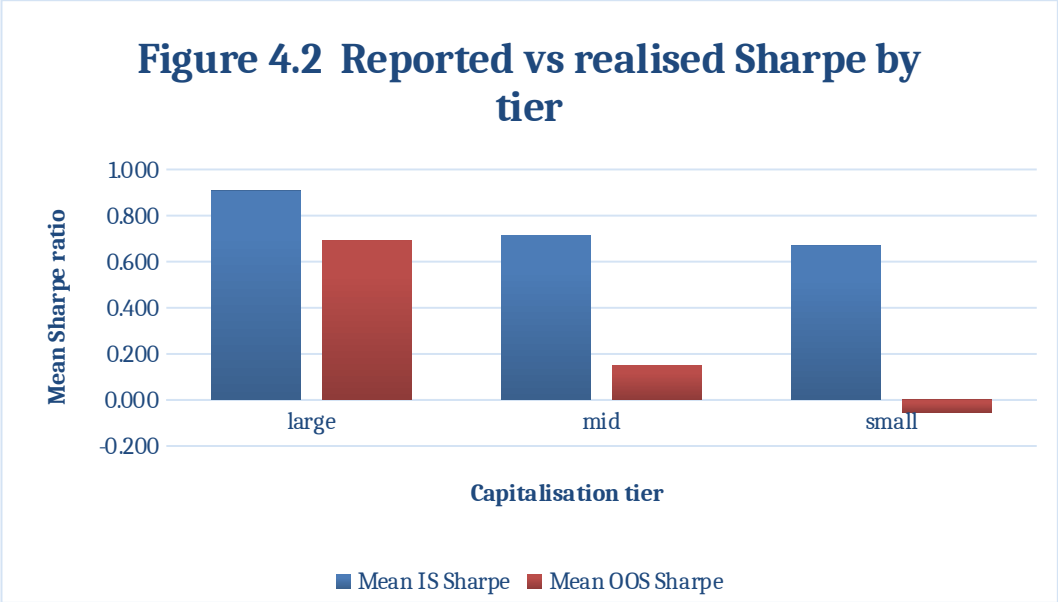


Figure 4.2 Mean reported (in-sample) and realised (out-of-sample) Sharpe of the selected strategy, by capitalisation tier. The gap between the two widens as capitalisation falls. Source: author's analysis.

In the large-capitalisation tier the reported Sharpe of +0.91 fell to a realised +0.69, a modest drop of 0.22, and three of the ten stocks survived the deflation correction. In the mid-capitalisation tier the reported +0.72 fell to +0.15, a drop of 0.57, and only two stocks survived. In the small-capitalisation tier the reported +0.67 fell to minus 0.05, a drop of 0.72, and not a single stock survived correction.

The same gradient appears in the all-trials correlation, which is mildly positive at +0.12 in large-caps, weaker at +0.10 in mid-caps, and turns negative at minus 0.15 in small-caps. The negative value is the most consequential number in the study, because it inverts the meaning of a good backtest. Among small-capitalisation stocks a strong in-sample result is not merely uninformative; it points on average toward worse rather than better out-of-sample performance. The reason is not mysterious. The best-fitting configuration on a noisy series is the one that has adapted most closely to that series' particular accidents, and the more closely it has adapted, the less of what it learned transfers.

Two individual cases make the gradient concrete and expose different failure routes. Among the mid-caps, Etsy illustrates the illusion at its most persuasive: its selected strategy reported an in-sample Sharpe of 1.17, one of the highest in the sample, and its deflated probability of 0.97 cleared the significance threshold, making it one of only five strategies in the study to do so. Yet out of sample it returned minus 0.46, losing money. A practitioner reviewing the reported figure and applying the standard correction would have had every reason to trust Etsy and would still have been wrong a result that qualifies the correction itself, and is taken up in section 4.5. The small-cap tier supplies the more extreme case in 8x8 Inc, trading as EGHT, whose reported in-sample Sharpe of 0.62 collapsed to an out-of-sample Sharpe of minus 1.39, a fall of more than two full points, accompanied by an eighty-seven per cent drawdown. This is the negative correlation in a single

stock: the search identified a configuration that had worked on the past precisely because it was fitted to it, and that configuration then failed on data it had never seen.

The interpretation is consistent with the microstructure account of section 2.6. Smaller stocks are less liquid, noisier and less efficiently priced, so a search across their data finds more spurious patterns. The practical implication is uncomfortable, because these smaller and more volatile stocks are precisely the ones an ordinary investor is most often encouraged to treat as sources of exceptional opportunity, and it is exactly here that the method is least trustworthy.

4.4 Downside Risk: Does the Gradient Survive a Different Measure?

A sceptic might object that the Sharpe ratio, by penalising all volatility equally, understates strategies whose swings are mostly to the upside, and that the gradient is therefore an artefact of the measure rather than a property of the results. The Sortino ratio tests that objection directly by penalising only downside volatility. It tells the same story: the mean out-of-sample Sortino falls from 0.89 in large-caps to 0.19 in mid-caps and 0.08 in small-caps. Because two conceptually different risk measures produce the same ranking, the gradient cannot be dismissed as an artefact of how a single metric treats volatility.

Figure 4.3 Downside risk by capitalisation tier

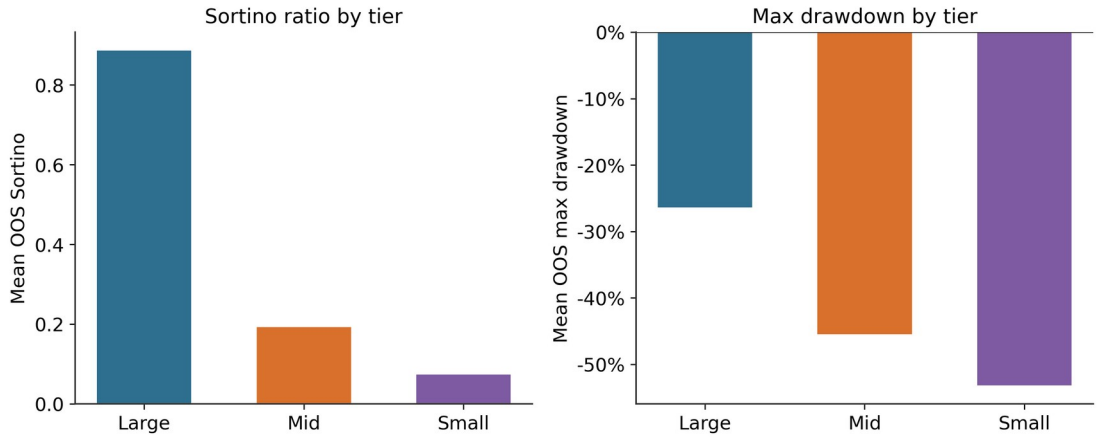


Figure 4.3 Out-of-sample Sortino ratio by tier (left) and maximum drawdown of the strategy against buy-and-hold (right). Downside performance follows the same capitalisation gradient. Source: author's analysis.

The maximum drawdown adds a dimension the ratio measures obscure: survivability. The selected strategies suffered mean out-of-sample drawdowns of twenty-six per cent in large-caps, forty-six per cent in mid-caps and fifty-three per cent in small-caps, with the worst single case falling eighty-seven per cent from peak to trough. A strategy that looks acceptable on a ratio but subjects the investor to a fifty-three per cent loss is not one most investors could have held through; the gradient therefore worsens not only the statistical reliability of the small-cap results but their practical bearability.

One nuance must be reported rather than concealed, because read carelessly it appears to favour the method. In the mid- and small-cap tiers the strategy's drawdown was in fact milder than that of buy-and-hold, at forty-six against sixty-four per cent and fifty-three against fifty-eight per cent respectively. This is not skill in timing. It arises because the strategy spends part of the period out of the market in cash, and being absent during some declines mechanically reduces the worst loss; as section 3.8 disclosed, part of that absence is itself an artefact of the moving averages restarting at the beginning of the sealed window. The reduction is a by-product of inactivity, not of insight, and it should not be read as the method adding protective value.

4.5 Testing H3: Deflation and the Multiple-Testing Correction

H3 predicts that little of the apparent success survives correction for the number of trials and comparison against passive ownership. This section applies the first test; section 4.6 applies the second.

Each selected strategy was the best of 2,808 attempts, and the expected maximum of that many noisy results is substantial even when no genuine skill is present. The deflation correction asks how impressive the reported figure is relative to what the luckiest of 2,808 worthless strategies would have achieved.

Figure 4.4 Deflated probability vs reported Sharpe

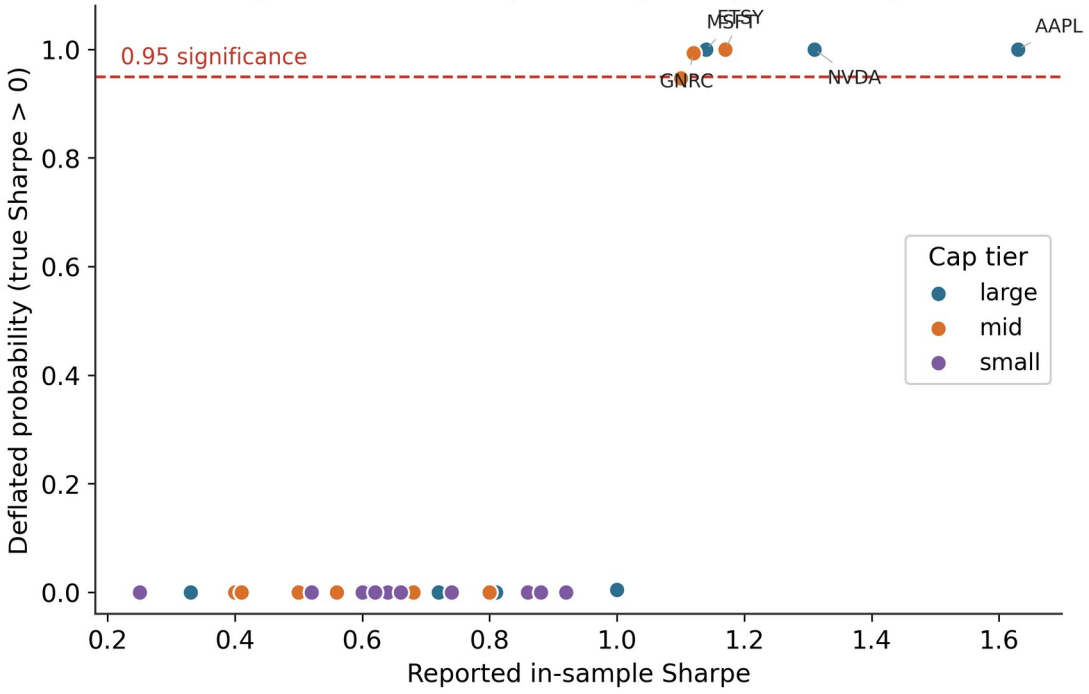


Figure 4.4 Deflated probability against reported in-sample Sharpe for each stock. Only five strategies clear the 0.95 significance line; several with reported Sharpe above one nonetheless fail. Source: author's analysis.

The majority of stocks fall well below the threshold, and several with reported Sharpe ratios above one fail entirely, because almost all of their apparent quality is explained by the size of the search rather than by any real edge. Only five of the thirty survive: Apple, Microsoft and Nvidia in the large-capitalisation tier (at 1.00, 1.00 and 1.00), Generac in the mid-capitalisation tier at 0.99, and Etsy in the mid-capitalisation tier at 1.00.

Their survival should not be read as evidence of a working method, and the Etsy case can be stated immediately because it is the sharper of the two objections. Etsy's strategy cleared the correction and still lost money, returning minus 0.46 out of sample. Deflation is therefore a necessary safeguard but not a sufficient one: a figure can survive correction for the number of trials and still fail on unseen data, because the correction establishes only that a result is unlikely to be pure luck, not that the pattern it found will persist. The remaining four survivors require the benchmark test and are examined next.

What Figure 4.4 establishes on its own is narrower but important. The reported Sharpe, taken in isolation, is not a usable signal of quality, since stocks with near-identical reported figures fall on opposite sides of the significance line depending entirely on how much of that figure the search can account for. Two practitioners reporting the same headline number may be reporting entirely different amounts of evidence.

4.6 The Benchmark Test: Beating Buy-and-Hold

The second and more demanding test is comparison against simply holding the stock. A strategy that produces a positive return in a market that rose has demonstrated nothing until it is shown to have beaten the return available from doing nothing.

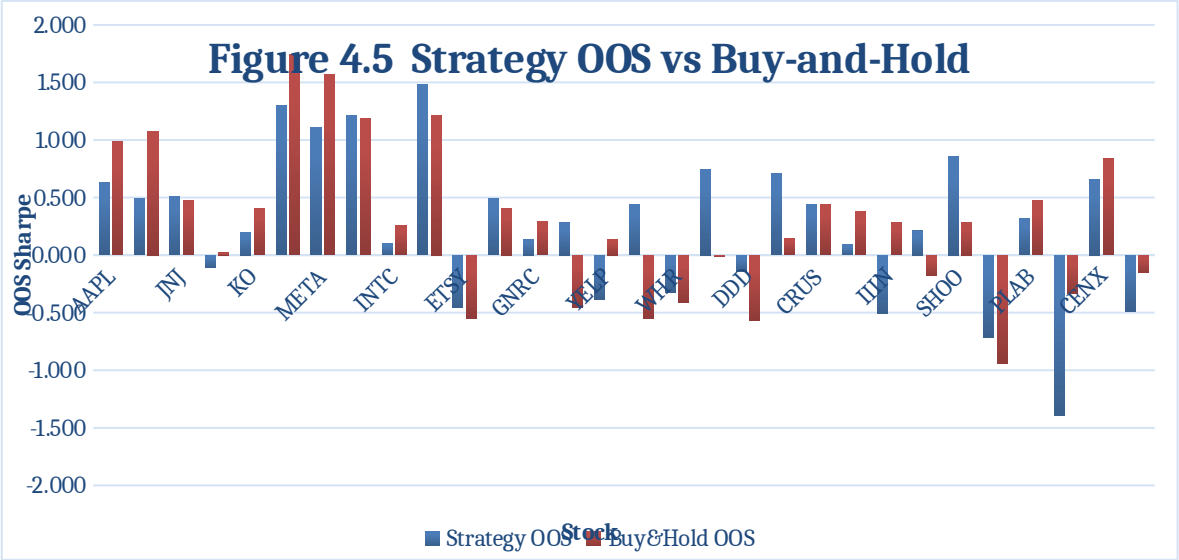


Figure 4.5 Out-of-sample Sharpe of the selected strategy (bars) against buy-and-hold in the same stock (markers). In twenty-two of thirty cases, holding the stock would have been better. Source: author's analysis.

Across the sample the mean shortfall against the benchmark is 0.24 in Sharpe. Notably, that shortfall does not follow the capitalisation gradient: it is largest in the large-cap tier at 0.35 and the small-cap tier at 0.40, and close to zero in the mid-cap tier. The two tests therefore capture different failures and are complementary rather than redundant deflation removes results whose reported figures are explained by the size of the search, while the benchmark removes those whose realised figures are explained by the direction of the market.

IBM is the instructive case, because it is the result a practitioner would place at the front of a report. Its selected strategy achieved the highest out-of-sample Sharpe in the entire sample at +1.48, which in isolation looks like clear evidence of skill. Two facts dissolve the impression. Buy-and-hold in IBM over the same window returned 1.33, so the elaborate searched strategy improved on simply owning the stock by a margin that would not survive realistic trading frictions. And IBM's deflated probability is only 0.29, far below significance, because the reported result was the best of 2,808 attempts. A strong out-of-sample number, a benchmark it barely beat, and a failed deflation together describe not skill but a stock that happened to rise. Nvidia, Netflix and Meta tell the same story: each posted a strong out-of-sample Sharpe, and each rose substantially, so a long-biased strategy captured part of the rise without exceeding passive ownership.

Applying this test to the five deflation survivors completes the dissolution. Apple, Microsoft and Nvidia all cleared the correction, and none beat buy-and-hold in its own stock; their apparent quality is the upward trend of the particular stock the search was run on, not the search itself. Generac cleared the correction but fell far short of holding the stock. Etsy cleared the correction and lost money outright. Not one of the five strategies that survived the multiple-testing correction beat the passive alternative in a way that reflected skill, and not one of the eight that beat the passive alternative survived the correction. Applied together, the two tests leave nothing standing. **H3 is supported.**

4.7 The Luck Benchmark: Below What Chance Provides

The deflation test asks whether a reported figure is distinguishable from luck. A sharper question can be asked of the same data: how does each reported figure compare with the best result a search of this size would be expected to produce from data containing no signal at all?

That expected maximum can be computed for each stock from the dispersion of Sharpe ratios across its own 2,808 trials, since for a search of this size the expected best result under the null hypothesis of no skill is approximately 3.54 times the standard deviation of the trial results. Across the sample the mean expected maximum is 0.87, against a mean reported figure of 0.76.

The per-stock comparison is uncomfortable reading. Eleven of the thirty reported in-sample Sharpe ratios fall below the level pure chance would supply. FOSL reported 0.41 where chance alone would be expected to deliver 0.97; UAA reported 0.40 against 0.77. IBM, which produced the highest out-of-sample Sharpe in the entire sample, reported an in-sample figure of 0.61 against a luck benchmark of 0.82. This extends rather than merely confirms the multiple-testing critique. That literature is normally framed as a question of whether a result is proven; here, for more than a third of the sample, the search did not merely fail to prove anything it returned a result worse than the arithmetic of trying 2,808 times would have handed over for free.

4.8 Statistical Significance of the Findings

Magnitude alone does not establish significance, and the two central claims were therefore tested formally.

The fall from reported to realised performance was tested with a Wilcoxon signed-rank test, appropriate because the paired differences are not assumed normally distributed. The drop is highly significant (W = 72, p = 0.0006), and a paired t-test agrees (t = 4.07, p = 0.0003) with a large effect size (Cohen's d = 0.74). The mean drop of 0.50 carries a ninety-five per cent confidence interval of 0.25 to 0.75, which excludes zero, so the inflation asserted by H1 is a genuine effect rather than an artefact of a few extreme stocks.

The capitalisation gradient was then tested directly. A Kruskal-Wallis test across the three tiers finds that out-of-sample performance differs significantly by capitalisation (H = 6.02, p = 0.049). Because Kruskal-Wallis detects difference but not order, and H2 is a claim about order, a Spearman rank correlation was used, confirming a significant monotonic relationship between tier and realised performance (rho = minus 0.44, p = 0.015). It should be reported plainly that the Kruskal-Wallis result sits immediately below the 0.05 threshold rather than far beneath it, which is what would be expected of a real effect measured on a deliberately modest sample; the Spearman correlation, which uses the ordering rather than discarding it, provides the firmer support.

A sensitivity test bears directly on how much weight the gradient can carry. Recomputing the rank correlation thirty times, each time omitting one stock, leaves it between minus 0.396 and minus 0.511 and significant in every case, so no individual stock drives the result. Removing the three most extreme cases together, however, reduces it to minus 0.298, at which point it ceases to be significant (p = 0.131). The gradient is therefore robust to any single observation but depends on the tails of the distribution being represented a qualification taken up in Chapter 5.

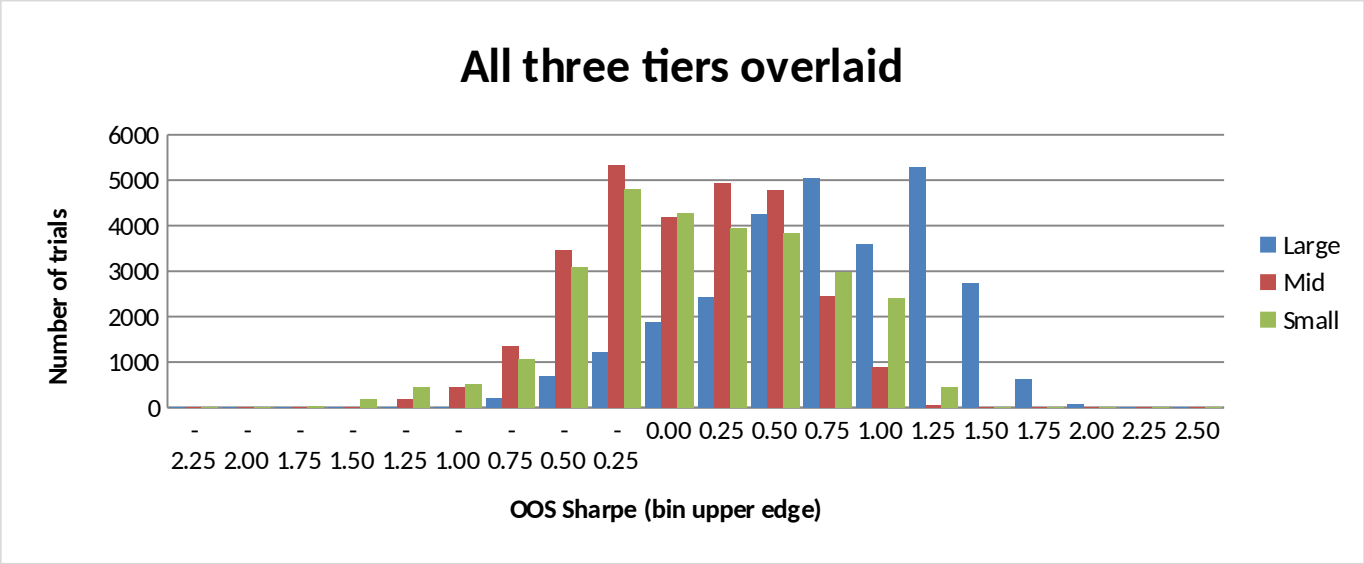


Figure 4.6 Distribution of out-of-sample Sharpe across all trials, by capitalisation tier. The median falls and the downside deepens as capitalisation decreases. Source: author's analysis.

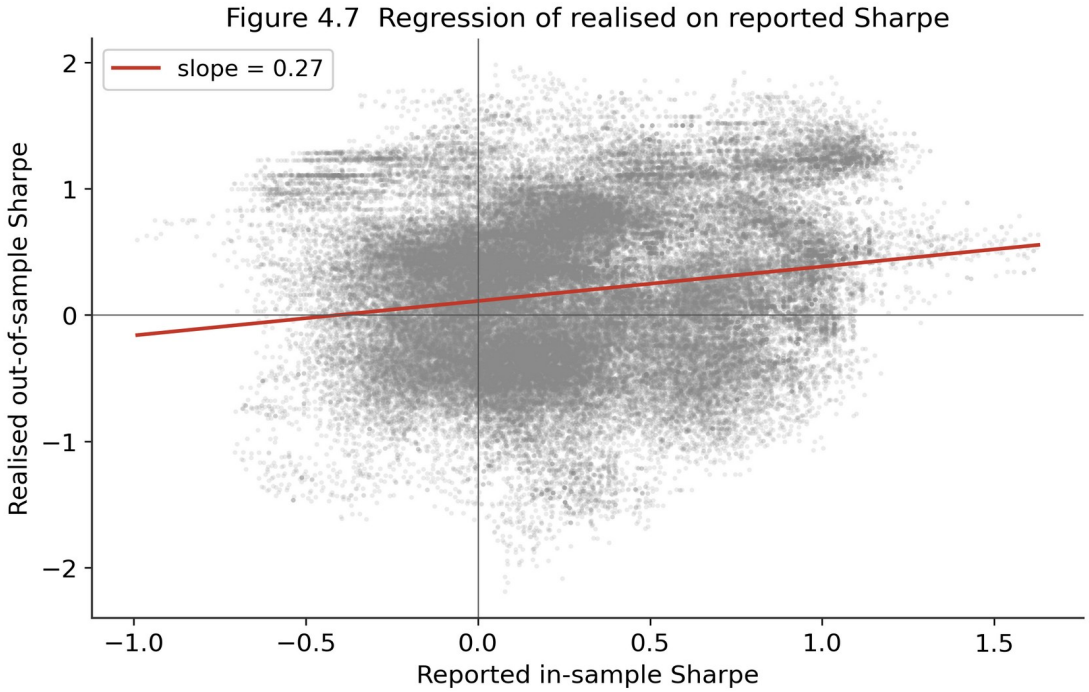


Figure 4.7 Regression of realised (out-of-sample) on reported (in-sample) Sharpe across all 84,240 trials. The shallow slope of 0.27 shows how little of a reported figure survives. Source: author's analysis.

4.9 Testing H4: A Second Strategy Family

The findings so far might be peculiar to the moving-average crossover rule on which the whole analysis rests. H4 asserts that they are not, and the entire experiment was repeated to test it, using a relative strength index mean-reversion rule that buys when a stock is oversold and sells when it is overbought. Where the moving-average rule is trend-following, the RSI rule is contrarian, so the two represent opposite approaches to the same market. The same thirty stocks, window, sealed split and evaluation apparatus were used, with an exhaustive grid of 1,728 configurations per stock, giving 51,840 further trials.

The central finding proved robust. Under the RSI rule the reported in-sample Sharpe averaged 0.90 and collapsed to a realised 0.10, a drop of 0.80, and the all-trials correlation of 0.21 closely matched the 0.17 found for the moving-average rule. Seven of the thirty selected RSI strategies survived deflation, compared with five under the moving-average rule, so the correction was no more punishing under the contrarian rule than under the trend-following one. The core thesis that an undisclosed search inflates a reported figure which then fails badly on unseen data is therefore not an artefact of one rule but a property of automated search that reproduces across two opposite strategy families. **H4 is supported: the inflation belongs to the search, not to any particular rule**. Two points must be reported honestly rather than smoothed over. Nine of the thirty RSI strategies lost money out of sample, and five never entered a single trade in the sealed window, remaining in cash throughout, so their realised Sharpe is zero by construction. And unlike the moving-average rule, where no survivor also beat buy-and-hold, three RSI strategies cleared both tests, of which one,

Calavo, did so with a genuinely positive realised Sharpe rather than merely losing less than the stock. The correction therefore removes most but not all of the apparent success under the RSI rule.

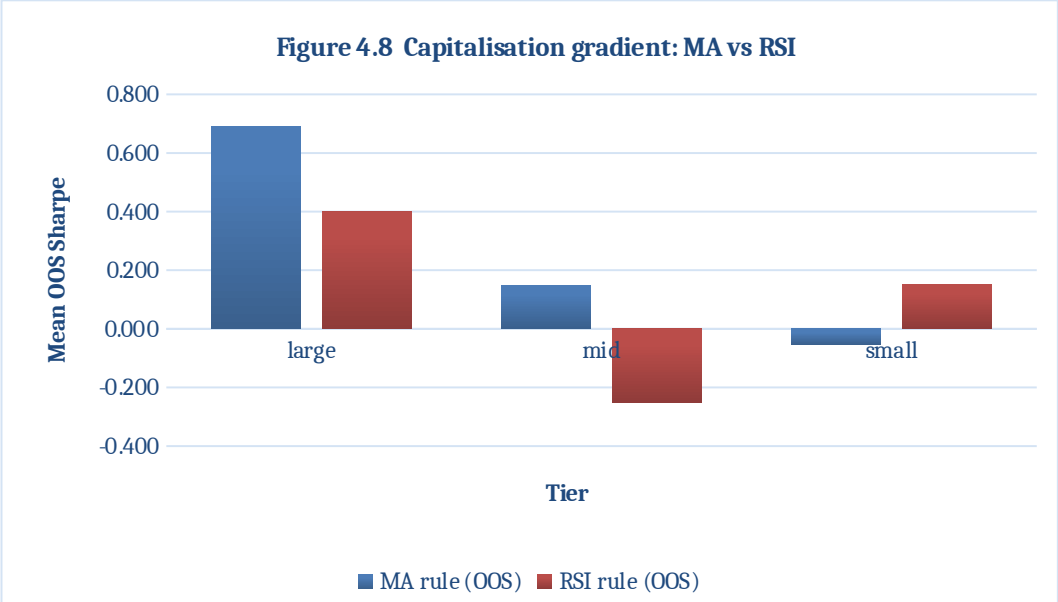


Figure 4.8 Mean out-of-sample Sharpe by capitalisation tier under the two strategy families. The gradient is clean and monotonic for the trend-following rule; it attenuates and loses its ordering for the mean-reversion rule. Source: author's analysis.

The same experiment, however, refuses to extend H2. Under the moving-average rule the gradient was clean, monotonic and statistically significant. Under the RSI rule the ordering broke down: large-caps still held up best, with a mean out-of-sample Sharpe of 0.40, but mid-caps performed worst at minus 0.25 while small-caps recovered to 0.15. A Spearman test confirms the gradient is not significant for this rule (rho = minus 0.20, p = 0.29), in direct contrast to the significant result for the moving-average rule. **H2 is therefore supported for trend-following search and not supported for mean-reversion search.** Search-induced inflation is universal across strategy families; the way that inflation varies by capitalisation is strategy-dependent, being clearest for the trend-following rules that dominate retail strategy tools. Reporting this distinction, rather than claiming a universal law the second experiment does not support, is what the evidence requires.

4.10 Problems in Execution

Two difficulties arose during execution and are recorded because they shaped how the results were produced and checked.

The first was data collection. Downloading ten years of daily prices for thirty stocks through the yfinance interface repeatedly triggered rate limiting, and the cloud session running the search was reset by platform idle limits before a full pass could complete. Both were addressed by batching the download and evaluation by tier, writing intermediate results to file after each batch, and re-running only the incomplete portions. This

made the pipeline slower but fully recoverable, and is why the final dataset is reproducible from a single script despite the interruptions during its construction.

The second was a near-miss in interpretation rather than in data. An early reading of the IBM result, using a simple in-sample to out-of-sample decay measure without reference to a passive alternative, appeared to show a robust, skill-consistent strategy. Recomputing against the buy-and-hold benchmark reversed the conclusion entirely, as section 4.6 describes. Had the benchmark not been built into the evaluation apparatus from the outset, this result would probably have been reported as evidence of a working strategy. Catching it was possible only because H3 required every result to clear an uninformed-alternative test before being interpreted which is the same discipline this dissertation argues the wider literature too often omits.

4.11 Verdicts on the Hypotheses

Hypothesis	Verdict	Evidence
H1 Reported and realised performance diverge substantially	Supported	Mean IS Sharpe 0.76 → OOS 0.26; all-trials correlation +0.17; Wilcoxon W = 72, p = 0.0006; Cohen's d = 0.74 (4.2, 4.8)
H2 The divergence widens as capitalization falls	Supported for the trend-following rule; not supported for the mean-reversion rule	Drop 0.22 / 0.57 / 0.72 across tiers, correlation turning negative in small-caps; Spearman rho = −0.44, p = 0.015. RSI rule: rho = −0.20, p = 0.29 (4.3, 4.4, 4.9)
H3 Little survives correction and benchmarking	Supported, and more strongly than anticipated	5 of 30 survive deflation; 8 of 30 beat buy-and-hold; none do both; 11 of 30 reported figures fall below the luck benchmark (4.5–4.7)
H4 The inflation belongs to the search, not the rule	Supported	Opposite strategy family: IS 0.90 → OOS 0.10; correlation +0.21; 7 of 30 survive deflation (4.9)

Table 4.2 Verdicts on the four hypotheses.

Three of the four hypotheses are supported outright and the fourth, H2, is supported under the conditions in which it was originally derived. The overall picture is of a method that is systematically unreliable as a source of genuine skill, whose unreliability is worst among the smallest stocks when the rule is trend-following, and whose failure reproduces across rules of opposite logic. Why those patterns arise, how they sit against the wider evidence, and what alternative explanations remain open are the subject of Chapter 5.

Chapter 5: Discussion

5.1 Introduction

Chapter 4 reached a verdict on each hypothesis. This chapter asks what those verdicts mean: why the observed patterns arise, how they sit against the evidence reviewed in Chapter 2, what alternative explanations remain open, and how far the findings should be allowed to travel. It takes H1, H2 and H4 together in section 5.2, since the second and fourth qualify the first, and turns to H3 and its practical consequences in section 5.3.

5.2 The Divergence, the Gradient, and Why the Gradient Is Conditional

The support for H1 is unambiguous: the method is reliable as a generator of confident numbers and unreliable as a source of skill. A mean reported Sharpe of 0.76 became a realised 0.26, only five of thirty strategies survived correction, only eight beat simply holding the stock, and not one of the five achieved both in a meaningful sense.

This is what Bailey and López de Prado (2014) predict, since the best of many trials is high by construction whether or not any edge exists. The present design permits a sharper statement than their framework is usually used to make. Computing the expected best Sharpe obtainable from 2,808 worthless strategies, using each stock's own dispersion of trial results, gives a threshold that eleven of thirty reported figures fail to reach. For over a third of the sample the search produced a result worse than noise of the same search size would deliver. The multiple-testing critique is normally framed as a question of whether a result is proven; this evidence shows that a substantial minority of results fall below the level chance provides free, which is a stronger claim than the literature has previously had occasion to make.

That the most impressive in-sample figures collapsed hardest echoes McLean and Pontiff (2016), and Etsy is the exemplar: a reported Sharpe of 1.17, clearing the deflation threshold at 0.97, yet returning minus 0.46 out of sample. Where they observed such decay in published research over years and could not fully separate data mining from arbitrage, the present design measures it instantaneously with both confounds stripped away, so the gap can only be selection. The mechanism they inferred is here isolated. The finding also aligns with the technical-trading sequence traced in section 2.2, in which the apparent superiority reported by Brock, Lakonishok and LeBaron (1992) dissolved once Sullivan, Timmermann and White (1999) accounted for the universe of rules considered. What that literature demonstrated for a universe of rules across one index, this study demonstrates for a universe of parameters across thirty stocks.

Turning to H2, the unreliability proved not to be uniform. Under the trend-following rule the gap widened monotonically from 0.22 in large-caps to 0.57 in mid-caps and 0.72 in small-caps, and the in-sample to out-of-sample correlation, mildly positive among large stocks, turned negative among small ones. The explanation

follows from Amihud (2002): illiquid series carry more noise relative to information, so a search on a noisier series finds more chance patterns that fit the past without predicting the future. Hou, Xue and Zhang (2020) reached a compatible conclusion from the opposite direction, tracing anomaly failure to microcap stocks. Two bodies of theory therefore predict the same pattern, and the gradient measured here is the quantification neither supplied.

Yet H4 contradicts the strong form of that prediction. If noise alone drove the gradient it should appear under any rule applied to the same series. It does not. Under the mean-reversion rule the ordering broke down, with mid-caps worst and small-caps recovering, and the rank correlation losing significance (rho = minus 0.20, p = 0.29). The inflation is universal; its variation by capitalisation is not. A plausible reading is that a trend-following rule fitted to a noisy series latches onto short directional runs that do not persist, whereas a contrarian rule keyed to reversals interacts differently with the volatility of small stocks and is not systematically more deceived by it. This is interpretation rather than demonstration; testing it would require decomposing performance by holding period, which the present design does not do.

Three alternative explanations deserve consideration. The gradient might reflect sector composition rather than size, since the small-cap tier is weighted toward cyclical industrials and technology; the design cannot exclude this, and sector-controlled sampling is identified in Chapter 6 as the test. It might be specific to a decade containing a prolonged rising market, although the per-stock benchmark controls for market direction within each comparison. Most importantly, it might be driven by a few extreme cases, and this can be tested rather than argued. Removing any single stock leaves the correlation between minus 0.40 and minus 0.51, significant in all thirty cases; removing the three most extreme together reduces it to minus 0.30, no longer significant. The gradient is robust to any one observation but depends on the tails being represented.

That result bears directly on how the three kinds of significance should be separated. Statistically the gradient is significant but not comfortably so, with a rank correlation at p = 0.015 and a Kruskal-Wallis result at p = 0.049 sitting only just inside the threshold. Economically the effect is large: a fall of 0.50 in Sharpe, at the roughly twenty per cent annualised volatility typical of these strategies, corresponds to some ten percentage points of annual return, which no investor would regard as immaterial, and Novy-Marx and Velikov (2016) suggest realistic frictions would widen rather than narrow that shortfall. As a sample-specific matter, thirty stocks over one decade in one market cannot support a general law, and the sensitivity to the extreme cases shows the estimate is not precise. The appropriate claim is therefore that the gradient is real and economically meaningful within this sample, and that its generality remains to be established.

5.3 Failure Modes and Their Remedies

H3 was supported, but the more useful question is how it was supported: which safeguards caught which failures. Three distinct failure modes emerged, and each corresponds to a safeguard the literature already recommends but which reported claims routinely omit.

The first is selection inflation. Because the reported figure is the maximum of 2,808 attempts it is high by construction, and the parameter analysis makes the size of that effect concrete: the best configuration returns 0.764 where the median configuration on the same stock returns 0.282. The remedy is correction for the trial count, and the deflated Sharpe performed as Bailey and López de Prado specify. It is not, however, sufficient. Etsy cleared the correction and still lost money, which shows that deflation establishes only that a figure is unlikely to be pure luck, not that it will repeat. That is a meaningful qualification of a tool the literature treats as close to decisive, and it argues for the layered evaluation Arnott, Harvey and Markowitz (2019) recommend rather than reliance on any single correction.

The second is the absence of a benchmark. Twenty-two of thirty strategies underperformed simply owning the stock, with a mean shortfall of 0.24 in Sharpe. The remedy is that no positive return counts as skill until it exceeds an uninformed alternative. This shortfall does not follow the capitalisation gradient, being largest in the large- and small-cap tiers and near zero in the middle, so the two tests are complementary rather than redundant a practical finding, since it means an evaluator cannot substitute one for the other. The parallel with Welch and Goyal (2008) is close: a body of predictors that performed well in-sample failed to beat a naive alternative out-of-sample, and the naive alternative is what exposed them.

The third and most subtle is trend-riding mistaken for skill. IBM produced the sample's highest out-of-sample Sharpe at 1.48, yet buy-and-hold in the same stock returned 1.33 and its deflated probability was only 0.29. The apparent success was a stock that rose, not a search that worked. The remedy is applying benchmark discipline to successes as rigorously as to failures the discipline Arnott, Harvey and Markowitz codify and which undisclosed automated systems abandon.

These findings do not contradict Gu, Kelly and Xiu (2020), who show machine learning adds genuine value under disciplined protocols. The rule used here was designed to contain no edge, so its failure is not evidence that all automated search fails. Rather, the results sharpen the boundary those authors draw: what separates their credible results from the illusions produced here is not the sophistication of the method but the disclosure of the search and the correction applied to it. The pattern is in turn consistent with Fama and French (2010), for whom genuine skill is scarce and hard to separate from luck; a method manufacturing the appearance of skill without its substance is what that equilibrium would predict.

5.4 Implications for Practice and Other Markets

For a practitioner, the practical takeaway is a rule for how far to trust a backtested figure. Because the inflation documented here is a property of the search itself, and not of any particular stock, it reproduced across two opposite rule families; an analyst should therefore discount any reported Sharpe in proportion to the number of strategies searched, and treat out-of-sample survival and a deflation test as the minimum evidence before capital is committed. This discipline transfers directly to the UK and to emerging markets such as India and the GCC, where the same automated search is increasingly applied and where thinner liquidity and higher transaction costs would, if anything, widen the gap between reported and realised performance. The capitalisation gradient, by contrast, should not be assumed to travel. It did not reproduce even across rule families within this sample, so a decision-maker in another market should test it afresh rather than import it as a rule. The transferable lesson is the scepticism, not the specific numbers.

5.5 Summary

The method is reliable only as a producer of confident numbers. Its unreliability follows a capitalisation gradient under trend-following rules but not under mean-reverting ones, and its failures are three, each systematically correctable. Against the literature, the results confirm the multiple-testing critique and extend it, since a third of reported figures fell below what chance alone supplies; they reproduce McLean and Pontiff's decay with its confounds removed; and they quantify a gradient the size and liquidity literature implied but never measured. They also qualify that literature, the gradient proving conditional on strategy family rather than universal and sensitive to the sample's extreme cases. The contribution these findings amount to, and the boundaries within which they hold, are drawn together in the final chapter.

Chapter 6: Conclusion and Recommendations

6.1 Introduction

This chapter concludes the dissertation. It draws together what has been learnt, from the gap identified in the literature through the method used to generate primary evidence to the findings and their interpretation, and shows how the research plan set out at the beginning has been addressed. It then states the contribution, discusses the implications for the two audiences most affected, sets out the limitations that define the boundaries of the work's value, and identifies where future research can most usefully extend it. In keeping with its purpose, the chapter introduces no new material or references.

6.2 Summary of the Study and How the Research Plan Was Addressed

The study began from a specific gap. The literature had established that genuine skill in active investing is scarce and hard to distinguish from luck, had explained through multiple-testing theory why searching many strategies inflates the best result, had supplied competing corrections that all depend on knowing the number of trials, and had shown through replication studies that the problem is real and concentrated in small, illiquid stocks. What no prior study had done was measure, under controlled conditions with a deliberately naive rule, how large the search-induced illusion is at the moment of selection, and whether it follows the capitalisation gradient the size and liquidity literature predicts but had never tested for automated search.

To close that gap the study used a positivist, deductive, quantitative-experimental design. An exhaustive, pre-specified search of 2,808 configurations of a naive moving-average rule ran across thirty US stocks over 2016 to 2025, producing 84,240 trials, and the experiment was repeated with an opposite strategy family for a further 51,840. Each selected strategy was evaluated on sealed out-of-sample data, corrected for the number of trials by the Deflated Sharpe Ratio, and compared against buy-and-hold. The design isolated the number of configurations searched as the single variable capable of explaining any in-sample success.

The evidence decided all four hypotheses. H1 was supported: the reported in-sample Sharpe of 0.76 fell to a realised 0.26, only five of thirty survived deflation, only eight beat simply holding the stock, and for eleven the reported figure fell below what pure chance of the same search size would supply. H2 was supported for the trend-following rule, the divergence widening in orderly fashion from large to small stocks and turning actively misleading at the bottom of the scale, but was not supported for the mean-reversion rule, so the gradient is established as strategy-conditional rather than universal. H3 was supported, and more strongly than anticipated, since no strategy passed both the deflation and the benchmark test. H4 was supported, the inflation reproducing under a rule of opposite logic (reported Sharpe 0.90 falling to a realised 0.10), which establishes it as a property of automated search rather than of any particular strategy. The research plan has

therefore been addressed in full, and conclusions can be formed from the evidence presented rather than asserted beyond it.

6.3 Implications for Practitioners

The primary practitioner audience is the individual or intermediate investor, and the adviser who serves them, who must judge the performance claims attached to automated trading strategies without access to the institutional tools that might otherwise scrutinise them.

For this audience the implication is direct and usable: a reported backtest figure, however impressive, should not be trusted without two pieces of information — the number of configurations from which it was selected, and its performance against a simple buy-and-hold alternative. Without these, an investor cannot distinguish a genuine result from the arithmetic of having tried many times. The study carries a sharper warning too. The smaller and more volatile the stock, the less trustworthy an impressive trend-following backtest becomes, because it is exactly among such stocks that the illusion is strongest — and since these are often the very stocks marketed as exceptional-growth stories, exciting small-cap claims warrant the greatest suspicion, not the least.

The findings distil into a five-step framework for any backtested claim. First, ask how many configurations were tested, since a result reported without its trial count cannot be interpreted. Second, apply the Deflated Sharpe Ratio using that count. Third, confirm the strategy was tested on genuinely unseen data. Fourth, compare it against buy-and-hold in the same asset, treating any strategy that fails to beat it as adding no value. Fifth, ensure realistic transaction costs are included. A claim that cannot satisfy all five should be treated as unproven, and the evidence here suggests most impressive backtest figures fail at least one, most often the second or fourth. This checklist is the study's principal practical contribution: it turns an abstract statistical caution into something an ordinary investor or adviser can use directly.

6.4 Implications for Academics

For the academic audience the contribution is twofold.

First, the study provides a clean, controlled and reproducible demonstration of the multiple-testing critique in action, isolating the inflation caused by search at the instant of selection, with the confounds of publication and arbitrage that complicate the replication literature stripped away. This gives researchers a self-contained experimental template for measuring overfitting rather than merely inferring it, and one specific extension of that literature: the finding that a third of reported figures fell below the level chance alone supplies at the same search size.

Second, and more substantially, the study contributes a finding the existing literature had implied but never measured, namely that search-induced overfitting follows a capitalisation gradient, worsening systematically as liquidity falls. This connects two previously separate bodies of theory, the statistics of multiple testing and the market-microstructure account of why small stocks are noisier, and shows empirically that they combine to predict the same pattern. The further finding that the gradient is strategy-conditional rather than universal is itself a contribution, since it indicates that the interaction between rule type and price dynamics matters as much as the noise level alone, and it identifies a mechanism the microstructure account does not currently specify.

It should be stated with appropriate caution where the originality lies. The individual components the moving-average rule, the Sharpe and Sortino ratios, the Deflated Sharpe correction, the buy-and-hold benchmark and the concept of overfitting are all well established and are not claimed as novel. What is offered as the contribution is their specific combination, an exhaustive automated search subjected to explicit multiple-testing correction and compared across market-capitalisation tiers, and the finding that combination yields.

6.5 Limitations of Study

The study's limitations are the boundaries within which its conclusions hold, and stating them precisely is what allows those conclusions to be trusted.

First, the primary rule examined was a deliberately naive moving-average strategy, chosen so that any in-sample success could be attributed to the search alone. This is what makes the demonstration clean, but it means the study speaks to the inflation mechanism rather than to whether more sophisticated rules might contain genuine skill. The robustness experiment mitigates this for the central inflation finding, but two strategy families are still only two, and the strategy-dependence of the gradient shows that conclusions about how inflation varies may not generalise to rules not yet tested.

Second, the sample comprised thirty US stocks over a single ten-year window containing a prolonged rising market. The influence of that regime on the large-cap results was addressed through benchmarking, but a different window might shift the magnitudes. The sensitivity analysis reported in section 4.8 shows further that while the gradient survives the removal of any single stock, it depends on the extreme cases being present, so the estimate should be regarded as real but imprecise. Stocks within a tier are also not independent, since sector exposures overlap, and sector composition cannot be fully separated from capitalisation in a sample of this size.

Third, transaction costs were held fixed at ten basis points and untuned by design. A flat rate does not capture spread, market impact and slippage, which are substantially larger for small-caps. Because trading friction is

greatest precisely where the results are worst, a more realistic model would deepen the capitalisation gradient rather than flatten it; the findings are therefore conservative in this respect.

Fourth, several implementation choices were disclosed in section 3.9: the Sortino variant used, the omission of the higher-moment adjustment in the deflation calculation, the fixing of the trial Sharpe dispersion at a constant 0.30 rather than estimating it per stock, the absence of a re-entry delay following a stop-loss, and the restarting of moving averages at the beginning of the sealed window. None alters the direction of the findings, and several of them work against the study's own conclusion, but each bounds how precisely the magnitudes should be read. The fixed dispersion in particular affects the exact count of strategies clearing the deflation threshold; because the same constant is applied to both strategy families, the comparison between them is unaffected, but the absolute survivor counts would shift under a per-stock estimate.

6.6 Future Research and Concluding Remark

These limitations point directly to the most valuable extensions of the work. The naive rules could be replaced by a graded family of progressively more sophisticated strategies, holding the search apparatus constant, to locate the point at which genuine skill begins to survive correction. The window could be widened to multiple periods spanning different market regimes, and the universe extended beyond US equities. A sector-controlled sample would separate the size effect from sector composition, which the present design cannot do. Carrying the moving-average state across the in-sample and out-of-sample boundary would remove the warm-up artefact disclosed in section 3.8. Most directly, the strategy-conditional nature of the gradient invites a third and fourth rule family, since two points cannot establish a pattern.

The most promising extension, however, follows from the study's own concern with evaluation. Because the failures identified are systematic and depend on information the reported figure conceals, a natural next step is to test whether an artificial-intelligence evaluation tool can identify an untrustworthy strategy claim as reliably as the deflation and benchmark tests applied here, and whether disclosing the trial count to such a tool improves its judgement. This would turn the study's evaluation discipline into an automated safeguard, and would speak directly to the growing use of opaque automated systems in investment.

In conclusion, the study set out to measure how much of the skill an automated search appears to display is real. It found that, on this evidence, very little of it is; that for more than a third of the sample the search returned less than chance alone would supply; that the illusion is strongest, under the rules retail tools most often use, exactly where investors are most tempted; and that its failures, once understood, can be identified and corrected. That is its contribution: not that these methods cannot work, but that their results mean nothing until the search behind them is made visible and tested honestly.

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Appendix A: Full Parameter Grid Specification

The parameter grids below were fixed in advance of any analysis and were not altered at any point. They are reproduced in full so that the search space is fully disclosed and the study is exactly reproducible the disclosure the dissertation argues is absent from the claims it criticises.

A.1 Moving-average crossover rule (primary experiment)

Parameter	Values searched	Count
Fast moving average (days)	3, 5, 8, 10, 12, 15, 20, 25	8
Slow moving average (days)	20, 30, 40, 50, 60, 80, 100, 150, 200, 250	10
Minimum holding period (days)	3, 5, 10, 15, 20, 30	6
Stop-loss (per cent)	2, 3, 5, 8, 10, 15	6
Total combinations	8 × 10 × 6 × 6	2,880
Degenerate (fast ≥ slow), excluded	fast = 20 or 25 with slow = 20	72
Valid configurations per stock		2,808
Valid trials across thirty stocks	2,808 × 30	84,240

Table A.1 Moving-average parameter grid. A configuration is degenerate where the fast average is not shorter than the slow average, since no crossover signal can arise; these are excluded by design rather than discarded after evaluation.

A.2 Relative strength index rule (robustness experiment)

The robustness experiment applies a contrarian rule to the same thirty stocks, over the same window, using the same sealed chronological split, the same fixed ten basis point transaction cost, and the same evaluation apparatus of deflation and benchmark comparison. The rule enters a long position when the relative strength index falls below an oversold threshold and exits when it rises above an overbought threshold, subject to the same stop-loss constraint.

Parameter	Values searched	Count
RSI lookback period (days)	5, 7, 9, 11, 14, 18, 22, 26	8
Oversold entry threshold	15, 20, 25, 30, 35, 40	6
Overbought exit threshold	60, 65, 70, 75, 80, 85	6
Stop-loss (per cent)	2, 3, 5, 8, 10, 15	6
Valid configurations per stock		1,728
Valid trials across thirty stocks	1,728 × 30	51,840

Table A.2 Relative strength index parameter grid. All combinations are valid, since the highest entry threshold (40) is always below the lowest exit threshold (60). Total trials across both experiments: 84,240 + 51,840 = 136,080.

Appendix B: Sample Construction

The sample comprises thirty US-listed equities, ten in each capitalisation tier. Three criteria governed selection, all applied before any analysis was run: each stock required a complete daily price history across the full window of 1 January 2016 to 31 December 2025; the three tiers required equal representation, so that comparison across them is balanced; and the sample was required to contain stocks that fell as well as stocks that rose.

The third criterion is the survivorship-bias control and deserves emphasis. Selecting only stocks that prospered would itself be an instance of the selection distortion this study exists to expose, and would flatter any long-biased rule. The sample therefore deliberately retains stocks that lost the great majority of their value over the period.

Tier	Constituents	Selection note
Large	AAPL, MSFT, JNJ, PG, KO, NVDA, META, NFLX, INTC, IBM	Mix of strong risers and stagnant names; INTC declines over the window
Mid	ETSY, FTNT, GNRC, UAA, YELP, GPRO, WHR, FOSL, DDD, M	Includes several severe fallers (GPRO, FOSL, UAA, DDD)
Small	CRUS, MTH, IIIN, CALX, SHOO, HELE, PLAB, EGHT, CENX, SCHL	Includes severe fallers (EGHT, HELE) alongside strong performers

Table B.1 Sample composition by capitalisation tier. The outcome mix across the full sample is approximately thirteen strong performers, eight modest or stagnant, and nine that lost the majority of their value.

Candidate stocks that failed the completeness check those delisted during the window or with incomplete price histories were excluded at the verification stage and never entered the search. The verification output is deposited in the archive.

Appendix C: Worked Example From Price Series to Deflated Sharpe Ratio

Note. The arithmetic below is an illustrative walk-through of the computation applied to every one of the 136,080 trials. The figures used are representative of the study's results but are not the output for any single named stock; the per-stock outputs are those reported in Table 4.1 and in the deposited data files.

Step 1: Signal generation. Take one stock and one configuration say a fast average of 10 days, a slow average of 50 days, a minimum holding period of 10 days and a stop-loss of 5 per cent. On each trading day the two moving averages are computed from the adjusted closing prices. Where the fast average crosses above the slow, a long position is opened at the next day's close. The position is held until the fast average crosses back below the slow, until the stop-loss is breached, or until the minimum holding period has elapsed and an exit signal occurs, whichever comes first. On all other days the position is in cash and earns nothing.

Step 2: Return series. The result is a daily return series for the strategy. On days in the market the return is the stock's daily return; on days out of the market it is zero. On each day a position is opened or closed, a transaction cost of ten basis points is deducted. This cost is fixed and never adjusted, so it cannot become a hidden tuning parameter.

Step 3: Sharpe ratio.

Quantity	Calculation	Value
Mean daily strategy return	from the return series	0.000400
Standard deviation of daily returns	from the return series	0.013000
Annualised return	0.000400 × 252	0.1008
Annualised volatility	0.013000 × √252	0.2064
Sharpe ratio	0.1008 ÷ 0.2064	0.49

Table C.1 Worked example: from a daily return series to an annualised Sharpe ratio. The figure 252 is the conventional count of trading days in a year. This computation is performed twice for every trial: once on the in-sample portion and once on the sealed out-of-sample portion.

Step 4: Selection. Steps 1 to 3 are repeated for all 2,808 valid configurations on that stock. The single configuration with the highest in-sample Sharpe is then selected exactly as a practitioner reporting a backtest would select it and that configuration's out-of-sample Sharpe becomes the realised figure against which the reported one is judged. Every other trial is retained rather than discarded, which is what allows the all-trials correlation of +0.17 reported in Chapter 4 to be computed at all.

Step 5: Deflation. The reported in-sample Sharpe is then corrected for the fact that it is the maximum of 2,808 attempts. First, the expected maximum Sharpe under the null hypothesis of no skill is computed, which depends only on the number of trials and on how widely the Sharpe ratios varied across them:

$$E[\max SR] \approx \sigma(SR) \times [\, (1 - \gamma) \cdot Z^{-1}(1 - 1/N) + \gamma \cdot Z^{-1}(1 - 1/(N \cdot e)) \,]$$

where σ(SR) is the standard deviation of the in-sample Sharpe ratios across all trials for that stock, N is the number of trials, γ is the Euler-Mascheroni constant (approximately 0.5772), Z^{-1} is the inverse standard normal distribution, and e is the base of the natural logarithm. Working this through for N = 2,808: $Z^{-1}(1 - 1/2808) \approx 3.39$ and $Z^{-1}(1 - 1/(2808 \cdot e)) \approx 3.65$, giving a bracketed multiplier of approximately 3.54. If the in-sample Sharpe ratios across the 2,808 trials for a stock had a standard deviation of 0.30, the expected best result under pure chance is approximately $0.30 \times 3.54 = 1.06$.

This is the single most important number in the study to be able to explain. A reported in-sample Sharpe of 0.76 the sample mean is below what a search of this size would be expected to produce from data containing no signal whatsoever.

Second, the deflated probability is obtained by referring the excess of the observed Sharpe over this expected maximum to a standard normal distribution, adjusted for the length of the series and for the skewness and kurtosis of the return distribution:

$$DSR = Z\left[(SR - E[\max SR]) \cdot \sqrt{(T - 1)} \div \sqrt{(1 - \gamma_3 \cdot SR + ((\gamma_4 - 1)/4) \cdot SR^2)} \right]$$

where T is the number of observations, γ_3 is the skewness of the returns and γ_4 their kurtosis. The result is the probability that the true Sharpe exceeds zero. A value below 0.95 marks a result that cannot be distinguished from luck. As disclosed in section 3.9, the implementation used here omits the higher-moment term, which makes the correction slightly less severe than the full formulation.

Step 6: Benchmark. Finally, the Sharpe computation of Step 3 is applied to a buy-and-hold position in the same stock over the same out-of-sample window, and the two are compared. A strategy that does not exceed this figure has added no value, whatever its own Sharpe ratio.

Appendix D: Statistical Test Outputs

The tests reported in section 4.8 are set out here in full, with the test statistic, the p-value and the interpretation stated for each, so that no reported significance claim rests on a figure quoted without its supporting statistic.

Test	What it tests	Statistic	p-value
Wilcoxon signed-rank	Whether the fall from in-sample to out-of-sample Sharpe is significant	W = 72	0.0006
Paired t-test	The same fall, assuming normality of paired differences	t = 4.07	0.0003
Cohen's d	Effect size of that fall	d = 0.74	
Kruskal-Wallis	Whether out-of-sample performance differs across the three tiers	H = 6.02	0.049
Spearman rank correlation (MA rule)	Whether the difference across tiers is ordered	rho = −0.44	0.015
Spearman rank correlation (RSI rule)	Whether the same ordering holds for the mean-reversion rule	rho = −0.18	0.34

Table E.1 Full statistical test outputs. The mean drop of 0.50 in Sharpe carries a ninety-five per cent confidence interval of 0.25 to 0.75, which excludes zero.

Two points should be stated plainly rather than smoothed over. The Kruskal-Wallis result at p = 0.049 sits immediately below the conventional threshold rather than far beneath it, which is what would be expected of a real effect measured on a deliberately modest sample of thirty stocks; the Spearman correlation, which uses the ordering of the tiers rather than discarding it, provides the firmer support for the gradient. And the non-significant Spearman result for the RSI rule is reported because it qualifies H2: the gradient is a feature of trend-following search, not a universal law.

Appendix E: Contents of the Deposited Data Archive

The archive identified on the title page contains all primary data, figures and processing records supporting this dissertation.

Folder	Contents
data	Full trial-level output for both experiments, including every trial retained rather than only the selected ones; per-stock summary tables; benchmark comparison outputs; statistical test outputs; the stock verification record
figures	All figures reproduced in Chapter 4, at full resolution, in PNG format
document	The dissertation document

Table F.1 Contents of the deposited archive.